

Topics on Electrical Engineering

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Part I

Magnetic Circuits

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Is it a fact - or have I dreamt it - that, by means of electricity, the world of matter has become a great nerve, vibrating thousands of miles in a breathless point of time?

(Nathaniel Hawthorne)

Chapter 1

Magnetic Circuits and Materials

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The primary objective of these series of lectures is to study devices used in the conversion between **electric** and **mechanical** energy. Of course, as this lecture is about electric drives,¹ emphasis is placed on **electromagnetic rotating machinery**, by means of which the majority of this energy conversion takes place.

¹On paper whereas mostly we will focus more on the machine aspect of drives.

The techniques developed here are generally applicable to a wide range of additional devices including **linear machines, actuators, and sensors**.

Although **NOT** an electromechanical-energy-conversion device by strictest definitions, the **transformer** is an important component of the overall energy-conversion process and is discussed in Chapter 2. The techniques developed for transformer analysis will form the basis for the ensuing discussion of electric machinery.

Practically, all transformers and electric machinery use **ferromagnetic** material² for shaping and directing the magnetic fields which act as the medium for transferring and converting energy, with Permanent Magnet (PM) materials also being widely used. Without these materials, practical implementations of most familiar electro-mechanical energy-conversion devices would **NOT** be possible.

²A property of certain materials, such as iron, which results in a significant, observable magnetic permeability, and in many cases, a significant magnetic **coercivity**, allowing the material to become a **permanent magnet**. Ferromagnetic materials are noticeably attracted to a magnet, which is a consequence of their substantial magnetic permeability [1].

The ability to analyse and describe systems containing these materials is vital for designing and understanding these devices.



1.1 An Overview on Magnetic Circuits

In this chapter we will develop the necessary tools for analysing of magnetic field systems and will provide a brief introduction to the properties of practical magnetic materials.

1.1.1 Maxwell's Equations

The complete, detailed solution for magnetic fields in most situations of practical engineering interest involves the solution of *Maxwell's equations*³ along with various constitutive relationships which describe material properties. While, in practice, **exact** solutions are often **unattainable**, various simplifying assumptions allow us to have approximate solutions.⁴

³A topic which becomes the center-point in **M.Sc Electrodynamics**.

We start our journey with the assumption that, for the systems treated in this lecture, the frequencies and sizes involved are such that the displacement-current (**D**) term in *Maxwell's equations* can be neglected. This term accounts for magnetic fields being produced in space by time-varying electric fields and is associated with electromagnetic radiation. Neglecting this term results in the magneto-quasistatic form of the relevant Maxwell's equations which relate magnetic fields to the currents which produce them [2]:

⁴Although exact analytical solutions cannot be obtained, numerical solutions are quite common and have become indispensable tools for analysis and design. However, their use contributes little to a fundamental understanding of the principles and basic performance of electric machines and therefore should be used after learning the fundamentals.

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{a} \quad (\text{Ampere's law}) \quad (1.1)$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0 \quad (\text{Gauss's law for magnetic fields}) \quad (1.2)$$

Let's have a look at these two (2) equations and try to understand what they mean;

1. Eq. (1.1) states that the line integral of the tangential component of the magnetic field intensity (**H**) around a closed contour (*C*) is equal to the total current passing through any surface (*S*) linking that contour. From Eq. (1.1) we see that the source of **H** is the current density **J**.
2. Eq. (1.2) states that the magnetic flux density (**B**) is conserved.⁵

⁵i.e., that no net flux enters or leaves a closed surface.

From these equations we see that the magnetic field quantities can be determined **solely** from the instantaneous values of the source currents and time variations of the magnetic fields follow directly from time variations of the sources.

A second simplifying assumption involves the concept of the **magnetic circuit**. The general solution for the magnetic field intensity (**H**) and the magnetic flux density (**B**) in a structure of complex geometry is extremely difficult. However, a three-dimensional field problem can often be reduced to what is essentially a one dimensional circuit equivalent, giving solutions of **within acceptable engineering accuracy** [3].

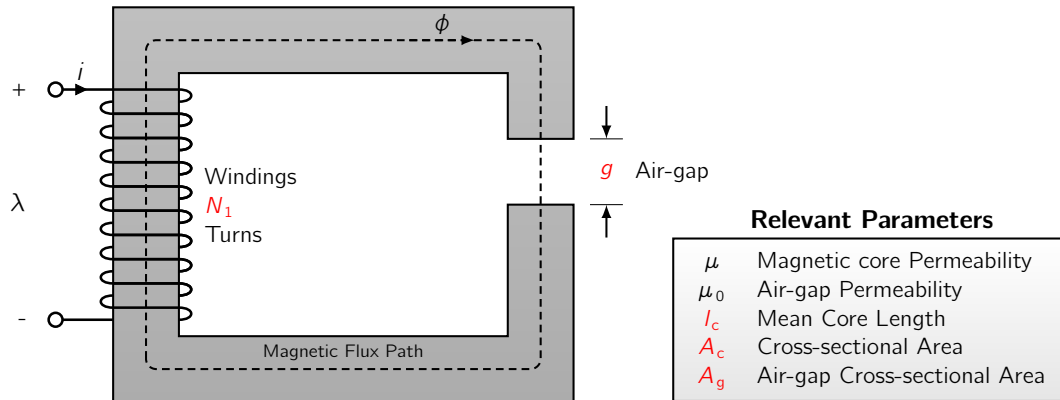


Figure 1.2

Simple magnetic circuit. λ is the winding flux linkage.

A magnetic circuit consists of a structure composed for the most part of **high-permeability** ($\mu \gg 0$) magnetic material. The presence of high-permeability material tends to cause magnetic flux to be **confined** to the paths defined by the structure, similar to currents being confined to the conductors of an electric circuit.

A simple example of a magnetic circuit is shown in **Fig. 1.2**. The core is assumed to be composed of magnetic material whose permeability is much greater than that of the surrounding air ($\mu \gg \mu_0$). The core is of **uniform** cross section and is excited by a winding of N turns carrying a current of i amperes. This winding produces a magnetic field in the core. Due to the high permeability of the magnetic core, an exact solution would show the magnetic flux is confined **almost entirely to the core**, the field lines follow the path defined by the core, and the flux density is **essentially uniform** over a cross section due to the cross-sectional area being uniform.

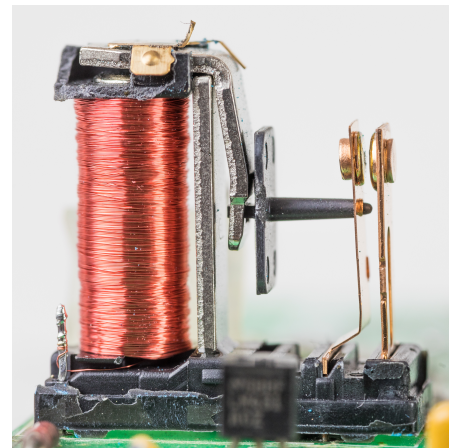


Figure 1.1

An example of a magnetic circuit in action. Relays are used in places where a signal needs to be cut-off at will. These electromechanical relays use magnetism to control the flow of electricity [4].

The magnetic field can be visualised in terms of flux lines which form closed loops interlinked with the winding.

As applied to the magnetic circuit of **Fig. 1.2**, the source of the magnetic field in the core is the ampere-turn product of Ni . In magnetic circuit terminology Ni is defined as the Magneto-motive Force (MMF) ($\mathcal{F}$) acting on the magnetic circuit.

Although **Fig. 1.2** shows only a single coil, transformers and most rotating machines have at least two (2) windings, and Ni therefore needs to be replaced by the **algebraic sum** of the ampere-turns of all the windings.

The magnetic flux (ϕ) crossing a surface ($\mathcal{S}$) is the surface integral of the normal component of

(B). Therefore

$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a} \quad (1.3)$$

In SI units,⁶ the unit of ϕ is the weber (Wb). As we can see, Eq. (1.2) states the net magnetic flux entering or leaving a closed surface, equal to the surface integral of $\mathbf{B}$ over that closed surface, is zero (0).

This is equivalent to saying all flux entering the surface enclosing a volume must leave that volume over some other portion of that surface as magnetic flux lines form closed loops.

These facts can be used to justify the assumption that the magnetic flux density is **uniform** across the cross section of a magnetic circuit such as the core of Fig. 1.2. In this case Eq. (1.3) reduces to the simple *scalar* equation:

$$\phi_c = B_c A_c,$$

where, ϕ_c is the core flux (Wb), B_c is the core flux density ($\text{Wb} \cdot \text{m}^{-2}$), and A_c is the cross-sectional area (m^2). Using Eq. (1.1), the relationship between the MMF acting on a magnetic circuit and the magnetic field intensity in that circuit is:

$$\mathcal{F} = Ni = \oint \mathbf{H} d\mathbf{l}. \quad (1.4)$$

The core dimensions are such that the path length of any flux line is close to the mean core length l_c . As a result, the line integral of Eq. (1.4) becomes simply the scalar product $H_c l_c$ of the magnitude of $\mathbf{H}$ and the mean flux path length l_c . Therefore, the relationship between the MMF and the magnetic field intensity can be written in magnetic circuit terminology as:

$$\mathcal{F} = Ni = H_c l_c$$

where H_c is **average magnitude** of $\mathbf{H}$ in the core.

The direction of H_c in the core can be found from the right-hand rule.⁷

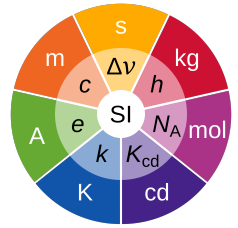
This can be stated in two (2) equivalent but slightly different ways:

- Imagine a current-carrying conductor held in the right hand with the thumb pointing in the direction of current flow. Here, the fingers then point in the direction of the magnetic field created by that current.
- Equivalently, if the coil in Fig. 1.2 is grasped in the right hand with the fingers pointing in the direction of the current, the thumb will point in the direction of the magnetic fields.

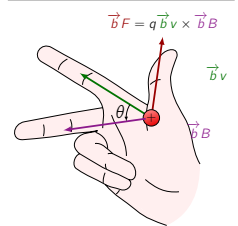
The relationship between the magnetic field intensity $\mathbf{H}$ and the magnetic flux density $\mathbf{B}$ is a property of the material in which the field exists. It is common to assume a **linear relationship**, which means:

$$\mathbf{B} = \mu \mathbf{H} \quad (1.5)$$

where μ is known as the **magnetic permeability**.



⁶The International System of Units, internationally known by the abbreviation SI (from French *Système international d'unités*), is the modern form of the metric system and the world's most widely used system of measurement. It is the only system of measurement with official status in nearly every country in the world, employed in science, technology, industry, and everyday commerce.



⁷A mnemonic, used to define the orientation of axes in three-dimensional space and to determine the direction of the cross product of two vectors, as well as to establish the direction of the force on a current-carrying conductor in a magnetic field.

In SI units, $\mathbf{H}$ is measured in units of $\text{A} \cdot \text{m}^{-1}$, $\mathbf{B}$ is in $\text{Wb} \cdot \text{m}^{-2}$, also known as teslas⁸ (T), and μ is in $\text{Wb} \cdot \text{A}^{-1} \cdot \text{Turn}^{-1} \cdot \text{m}^{-1}$, or equivalently ($\text{H} \cdot \text{m}^{-1}$). In SI units, the permeability of free space is $\mu_0 = 4\pi \times 10^{-7} \text{ H} \cdot \text{m}^{-1}$ and is a **physical constant**.

The permeability of linear magnetic material can be expressed in terms of its **relative permeability**, its value relative to that of free space, or $\mu = \mu_r \mu_0$. Typical values of μ_r range from $2,000 \times 10^3$ to 80,000 for materials used in transformers and rotating machines.

For now we will assume μ_r is a **known** constant, although it actually varies appreciably with the magnitude of the magnetic flux density.

Transformers are wound on closed cores similar to one we have seen in **Fig. 1.2**. However, energy conversion devices which incorporate a moving element must have air gaps in their magnetic circuits which is also shown in **Fig. 1.2**. When the air-gap length (g) is much smaller than the dimensions of the adjacent core faces, the magnetic flux will follow the path defined by the core and the air gap and the techniques of magnetic-circuit analysis can be used.

If the air-gap length were to become excessively large, the flux will be observed to **leak out** of the sides of the air gap and the techniques of magnetic-circuit analysis will no longer be strictly applicable. Therefore, provided the air-gap length g is sufficiently small, the configuration of **Fig. 1.2** can be analysed as a magnetic circuit with two (2) series components both carrying the same flux ϕ :

- a magnetic core of permeability μ , cross-sectional area A_c and mean length l_c , and
- an air gap of permeability μ_0 , cross-sectional area A_g and length g .

In the core, the magnetic field density is approximated to be:

$$B_c = \frac{\phi}{A_c} \quad (1.6)$$

and in the air gap, the value is:

$$B_g = \frac{\phi}{A_g} \quad (1.7)$$

If we apply Eq. (1.4) to aforementioned equations, this gives:

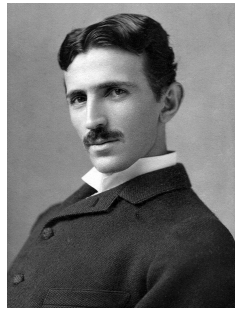
$$\mathcal{F} = H_c l_c + H_g g \quad (1.8)$$

and using the **linear B-H** relationship⁹ of Eq. (1.5) gives

$$\mathcal{F} = \frac{B_c}{\mu} l_c + \frac{B_g}{\mu_0} g \quad (1.9)$$

Here the $\mathcal{F} = Ni$ is the MMF applied to the magnetic circuit. From Eq. (1.8) we can see a portion of the MMF, namely $F_c = H_c l_c$, is required to produce magnetic field in the core whereas the remainder, $F_g = H_g g$ produces magnetic field in the air gap.

For practical applications, B_c and H_c are related **NOT** only to constant permeability μ .



⁸Nikola Tesla
(1856 - 1943)

A Serbian-American engineer, futurist, and inventor. He is known for his contributions to the design of the modern alternating current (AC) electricity supply system. The SI unit is magnetic flux density is named after him.

⁹This is, again, just an assumption of simplification.

In fact, B_c is often a **non-linear**, multi-valued function of H_c . Therefore, although Eq. (1.8) continues to hold, it does not lead directly to a simple expression relating the MMF and the flux densities, such as that of Eq. (1.9).

Instead, the specifics of the nonlinear $B_c - H_c$ relation must be used, either graphically or analytically. However, in many cases, the concept of constant material permeability gives results of acceptable engineering accuracy and is frequently used.

Using both Eq. (1.6) and Eq. (1.7), Eq. (1.9) can be rewritten in terms of the flux ϕ_c as:

$$\mathcal{F} = \phi \left(\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g} \right) \quad (1.10)$$

The terms that multiply the flux in Eq. (1.10), are known as the reluctance¹⁰ ($\mathcal{R}$) of the core and air gap, respectively,

$$\begin{aligned} \mathcal{R}_c &= \frac{l_c}{\mu A_c} \\ \mathcal{R}_g &= \frac{g}{\mu_0 A_g} \end{aligned} \quad (1.11)$$

and therefore we arrive at:

$$\mathcal{F} = \phi(\mathcal{R}_c + \mathcal{R}_g) \quad (1.12)$$

Finally, Eq. (1.12) can be inverted to solve for the flux

$$\begin{aligned} \phi &= \frac{\mathcal{F}}{\mathcal{R}_c + \mathcal{R}_g} \\ \phi &= \frac{\mathcal{F}}{\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g}} \end{aligned} \quad (1.13)$$

In general, for any magnetic circuit of total reluctance ($\mathcal{R}[tot]$), the flux can be found as

$$\phi = \frac{\mathcal{F}}{\mathcal{P}} \quad (1.14)$$

The term which multiplies the MMF is known as the permeance¹¹ ($\mathcal{P}$) and is the inverse of the reluctance. With that in mind, the total permeance of a magnetic circuit is:

$$\mathcal{P}_{tot} = \frac{1}{\mathcal{R}_{tot}}$$

As can be seen, Eq. (1.12) and Eq. (1.13) are analogous to the relationships between the current and voltage in an electric circuit.

This analogy is illustrated in **Fig. 1.3**. In the figure (a) shows an electric circuit in which a voltage V drives a current I through resistors R_1 and R_2 . (b) shows the schematic equivalent representation of the magnetic circuit of **Fig. 1.2**. Here we see that the MMF $\mathcal{F}$, analogous to voltage in the electric circuit, drives a flux ϕ , analogous to the current in the electric circuit, through the combination of the reluctances of the core $\mathcal{R}[c]$ and the air gap $\mathcal{R}[g]$. This analogy between the solution of

¹⁰A concept used in the analysis of magnetic circuits and defined as the ratio of MMF to magnetic flux. It represents the opposition to magnetic flux, and depends on the geometry and composition of an object.

It is analogous to electrical resistance in an electrical circuit in that resistance is a measure of the opposition to the electric current. The definition of magnetic reluctance is analogous to Ohm's law in this respect. However, magnetic flux passing through a reluctance does **NOT** give rise to dissipation of heat as it does for current through a resistance.

¹¹The inverse of reluctance and is a measure of the quantity of magnetic flux for a number of current-turns. A magnetic circuit almost acts as though the flux is conducted, therefore permeance is larger for large cross-sections of a material and smaller for smaller cross section lengths. This concept is analogous to electrical conductance in the electric circuit.

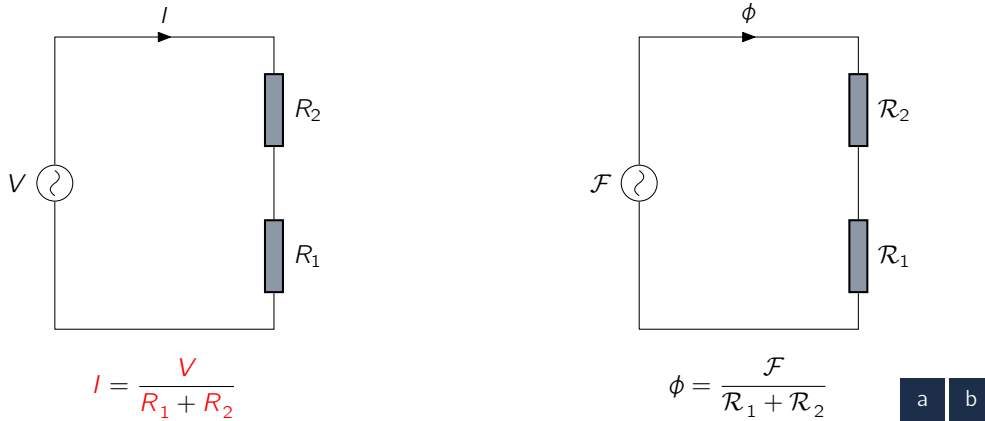


Figure 1.3

A visual comparison to show the conceptual similarities between an electric and magnetic circuit.

electric and magnetic circuits can often be exploited to produce simple solutions for the fluxes in magnetic circuits of considerable complexity.

The fraction of the MMF required to drive flux through each portion of the magnetic circuit, commonly referred to as the **MMF drop** across that portion of the magnetic circuit, varies in proportion to its reluctance.¹²

Consider the magnetic circuit of **Fig. 1.2**. From Eq. (1.11) we see that high material permeability can result in low core reluctance, which can often be made much smaller than that of the air gap. In this case, the reluctance of the core can be neglected and the flux can be found from Eq. (1.13) in terms of $\mathcal{F}$ and the air-gap properties alone:

$$\phi \approx \frac{\mathcal{F}}{\mathcal{R}_s} = \frac{\mathcal{F}\mu_0 A_g}{g} = Ni \frac{\mu_0 A_g}{g} \quad (1.15)$$

¹² This is directly analogous to the voltage drop across a resistive element in an electric circuit.

Understanding the Assumptions

1.1 Information

While we treat permeability as constant here, it is worth remembering that practical magnetic materials have permeabilities which are not constant but vary with the flux level. However, from Eq. (1.11) to Eq. (1.13) we see that as long as this permeability remains sufficiently large, its variation will not significantly affect the performance of a magnetic circuit in which the dominant reluctance is that of an air gap.

In practical systems, the magnetic field lines “fringe” outward somewhat as they cross the air gap. Provided this fringing effect is **NOT** excessive, the magnetic-circuit concept remains applicable. The effect of these fringing fields is to increase the effective cross-sectional area A_g of the air gap. Various empirical methods have been developed to account for this effect.

A correction for fringing fields in short air gaps can be made by adding the gap length to each of the two dimensions making up its cross-sectional area. For simplicity, we shall ignore fringing effects.

1.2 Flux Linkage, Inductance, and Energy

When a magnetic field varies with time, an electric field is produced in space as per Faraday's¹³ law:

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}$$

Eq. (1.16) states:

The line integral of the electric field intensity ($\mathbf{E}$) around a closed contour ($\mathcal{C}$) is equal to the time rate of change of the magnetic flux linking, passing through, that contour.

In magnetic structures with windings of high electrical conductivity, such as in **Fig. 1.2**, it can be shown, the $\mathbf{E}$ field in the wire is extremely small and can be neglected, so that the Left Hand Side (LHS) of Eq. (1.16) reduces to the negative of the induced voltage (e) at the winding terminals. In addition, the flux on the Right Hand Side (RHS) of Eq. (1.16) is dominated by the core flux ϕ_c . Since the winding, and therefore the contour $\mathcal{C}$, links the core flux N times, Eq. (1.16) reduces to:

$$e = N \frac{d\phi}{dt} = \frac{d\lambda}{dt} \quad (1.16)$$

where λ is the flux linkage of the winding and is defined as:

$$\lambda = N\phi \quad (1.17)$$

Flux linkage is measured in units of Wb (or equivalently Wb · Turn). The symbol ϕ is used to indicate the **instantaneous value** of a time-varying flux. In general the flux linkage of a coil is equal to the **surface integral** of the **normal component of the magnetic flux density** integrated over any surface spanned by that coil.

Direction of the induced voltage (e) is defined by Eq. (1.16) so if the winding terminals were **short-circuited**, a current would flow in such a direction as to **oppose the change of flux linkage**.

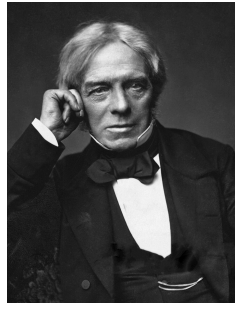
For a magnetic circuit composed of magnetic material of constant magnetic permeability or which includes a dominating air gap, the relationship between λ and i will be **linear** and we can define the inductance L as:

$$L = \frac{\lambda}{i} \quad (1.18)$$

Substitution of Eq. (1.4), Eq. (1.14) and Eq. (1.17) into Eq. (1.18) gives:

$$L = \frac{N^2}{\mathcal{R}_{\text{tot}}}$$

From which we see that the inductance of a winding in a magnetic circuit is proportional to the square of the turns and inversely proportional to the reluctance of the magnetic circuit associated with that winding.



¹³Michael Faraday
(1791 - 1867)

An English chemist and physicist who contributed to the study of electro-chemistry and electromagnetism. His main discoveries include the principles underlying electromagnetic induction, diamagnetism, and electrolysis. Although Faraday received little formal education, as a self-made man, he was one of the most influential scientists in history [5]. It was by his research on the magnetic field around a conductor carrying a direct current Faraday established the concept of the electromagnetic field in physics.

Faraday also established magnetism could affect rays of light and that there was an underlying relationship between the two phenomena. He similarly discovered the principles of electromagnetic induction, diamagnetism, and the laws of electrolysis. His inventions of electromagnetic rotary devices formed the foundation of electric motor technology, and it was largely due to his efforts that electricity became practical for use in technology. The SI unit of capacitance, the farad, is named after him.

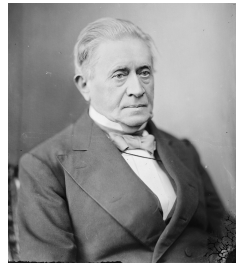
As an example, from Eq. (1.15), under the assumption that the reluctance of the core is negligible as compared to that of the air gap, the inductance of the winding in Fig. 1.2 is equal to

$$L = \frac{N^2}{(g/\mu_0 A_g)} = \frac{N^2 \mu_0 A_g}{g} \quad (1.19)$$

Inductance is measured in henrys¹⁴ (H) or weber-turns per ampere.

Eq. (1.19) shows the dimensional form of expressions for inductance;

Inductance is proportional to the square of the number of turns, to a magnetic permeability, and to a cross-sectional area and is inversely proportional to a length.



¹⁴ Joseph Henry
(1797 - 1878)

An American physicist and inventor who served as the first secretary of the Smithsonian Institution. While building electromagnets, discovered self-inductance. He also discovered mutual inductance independently of Michael Faraday, though Faraday was the first to make the discovery and publish his results and work on the electromagnetic relay was the basis of the practical electrical telegraph. In his honour, the SI unit of inductance is named the henry.

Requirements for Defining Inductance

1.2 Information

It must be emphasised that, strictly speaking, the concept of inductance requires a **linear relationship** between flux and MMF. Therefore, it cannot be rigorously applied in situations where the nonlinear characteristics of magnetic materials, dominate the magnetic system.

However, for many situations of practical interest, the reluctance of the system is dominated by that of an air gap, which is of course linear, and the non-linear effects of the magnetic material can be ignored.

In other cases it may be perfectly acceptable to assume an average value of magnetic permeability for the core material and to calculate a corresponding average inductance which can be used for calculations of reasonable engineering accuracy.

Plotting Permeability

1.1 Exercise

Using Python, plot the inductance of the magnetic circuit given in (TutBook) Ex. ?? as a function of core permeability over the range $100 \leq \mu_r \leq 100000$.

The script for the plot is as follows:

```
#+begin_src python
import numpy as np
import matplotlib.pyplot as plt

mu0 = 4 * np.pi * 1e-7      # (H/m) permeability of free space
lc = 0.3                    # (m) length
Ac = 9e-4                   # (m2) cross-sectional area
Ag = 9e-4                   # (m2) cross-sectional area
g = 5e-4                    # (m) air gap length
N = 500                     # (-) number of turns
Bc = 1                      # (T) Core magnetic field density

Rel_g = (g) / (mu0 * Ag)    # (Ampere-Turns/Weber) air-gap reluctance

mur = np.zeros(1000)       # Relative Permeability empty array for looping

for i in range(len(mur)):
```

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```

mur[i] = 100 + (100000 - 100) * (i - 1)/1000

Rel_c = (lc) / (mur * mu0 * Ac) # (Ampere-Turns/Weber) core reluctance
R_tot = Rel_c + Rel_g          # (Ampere-Turns/Weber) total reluctance

L = N**2 / R_tot               # (H) Inductance

# -- Plotting
plt.xlabel('Core Relative Permeability ($\mu_r$) [H/m]')
plt.ylabel('Inductance [H]')
plt.xlim(0, 0.8e5)
plt.ylim(0, 1)
plt.title("Core Relative Permeability vs. Inductance Graph", color='#c6d0f5')
plt.plot(mur, L, '#ef9f76', linewidth=5)
fig.tight_layout()

```

Please observe the resultant plot. The figure clearly confirms that, for the magnetic circuit of this example, the inductance is quite **insensitive** to relative permeability until the relative permeability drops to on the order of 1000. Therefore, as long as the effective relative permeability of the core is "large", which in this case greater than 1000, any non-linearities in the properties of the core material will have little effect on the terminal properties of the inductor.

Fig. 1.4 shows a magnetic circuit with an air gap and two (2) windings. In this case please observe the MMF acting on the magnetic circuit is given by the **total ampere-turns** acting on the magnetic circuit¹⁵ and that the reference directions for the currents have been chosen to produce flux in the same direction.

¹⁵i.e., the net ampere turns of both windings.

The total MMF is therefore calculated to be:

$$\mathcal{F} = N_1 i_1 + N_2 i_2$$

and from Eq. (1.15), with the reluctance of the core neglected and assuming $A_c = A_g$, the core flux

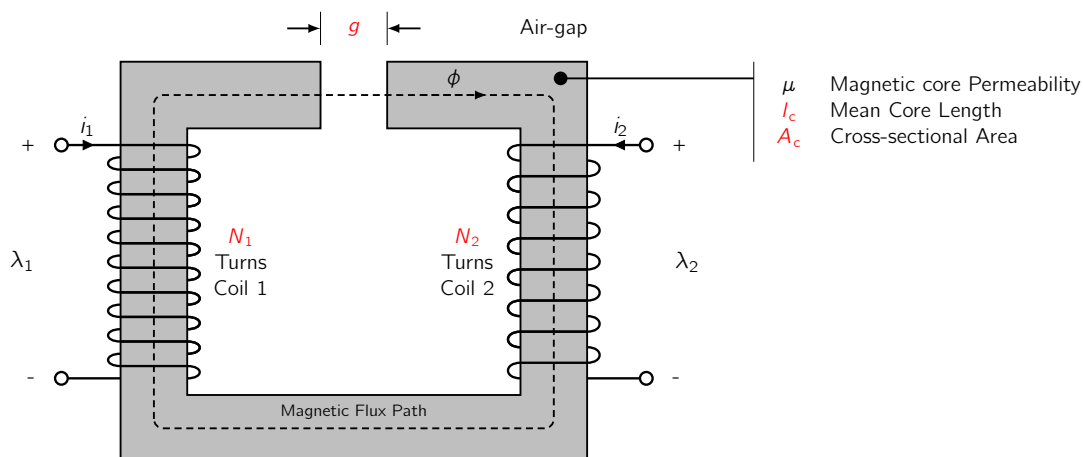


Figure 1.4: An example of a magnetic circuit, excited with two windings.

ϕ is:

$$\phi = (N_1 i_1 + N_2 i_2) \frac{\mu_0 A_c}{g} \quad (1.20)$$

In Eq. (1.20), ϕ is the resultant core flux produced by the total MMF of the two windings.

It is this resultant ϕ which determines the operating point of the core material.

If Eq. (1.20) is broken up into individual currents, the resultant flux linkages of coil 1 can be expressed as

$$\lambda_1 = N_1 \phi = N_1^2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_2$$

which can be written in a simpler form as:

$$\lambda_1 = L_{11} i_1 + L_{12} i_2$$

where:

$$L_{11} = N_1^2 \frac{\mu_0 A_c}{g}$$

is the **self-inductance** of Coil 1, and $L_{11} i_1$ is the flux linkage of Coil 1 due to its own current i_1 .

The **mutual inductance** between coils 1 and 2 is:

$$L_{12} = N_1 N_2 \frac{\mu_0 A_c}{g}$$

and $L_{12} i_2$ is the flux linkage of coil 1 due to current i_2 in the other coil. Similarly, the flux linkage of coil 2 is

$$\lambda_2 = N_2 \phi = N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_2^2 \left(\frac{\mu_0 A_c}{g} \right) i_2$$

or

$$\lambda_2 = L_{21} i_1 + L_{22} i_2$$

where $L_{21} = L_{12}$ is the mutual inductance and

$$L_{22} = N_2^2 \frac{\mu_0 A_c}{g}$$

is the self-inductance of coil 2.

It is important to note that the resolution of the resultant flux linkages into the components produced by i_1 and i_2 is based on superposition of the individual effects and therefore implies a linear flux-MMF relationship.¹⁶

¹⁶characteristic of materials of constant permeability.

Substitution of Eq. (1.18) in Eq. (1.16) gives:

$$e = \frac{d(Li)}{dt} \quad (1.21)$$

for a magnetic circuit with a single winding. For a static magnetic circuit, the inductance is fixed,¹⁷ and this equation reduces to the familiar circuit-theory form:

¹⁷assuming that material non-linearities do not cause the inductance to vary

$$e = L \frac{di}{dt}$$

However, in electromechanical energy conversion devices, inductances are often time-varying, and Eq. (1.21) must be written as

$$e = L \frac{di}{dt} + i \frac{dL}{dt}$$

Note that in situations with multiple windings, the total flux linkage of each winding must be used in Eq. (1.16) to find the winding-terminal voltage. The power at the terminals of a winding on a magnetic circuit is a measure of the rate of energy flow into the circuit through that particular winding. The **power**, p , is determined from the product of the voltage and the current

$$p = ie = i \frac{d\lambda}{dt} :$$

and its unit is watts (W), or ($\text{J} \cdot \text{s}^{-1}$). Therefore, the change in magnetic stored **energy** ΔW in the magnetic circuit in the time interval t_1 to t_2 is

$$\Delta W = \int_0^{t_2} p dt = \int_0^{\lambda_2} i d\lambda$$

In SI units, the magnetic stored energy W is measured in joules (J). For a single-winding system of constant inductance, the change in magnetic stored energy as the flux level is changed from λ_1 to λ_2 can be written as

$$\Delta W = \int_{\lambda_1}^{\lambda_2} i d\lambda = \int_{\lambda_1}^{\lambda_2} \frac{\lambda}{L} d\lambda = \frac{1}{2L} (\lambda_2^2 - \lambda_1^2) \quad (1.22)$$

The total magnetic stored energy at any given value of λ can be found from setting λ_1 equal to zero:

$$W = \frac{1}{2L} \lambda^2 = \frac{L}{2} i^2$$

The First Electrical Motor

Before their modern versions, motors using electrostatic forces were investigated. The first electric motors were simple electrostatic devices described in experiments by **Andrew Gordon** and **Benjamin Franklin** in the 1740s [7]. However, due to the difficulty of generating the high voltages they required, electrostatic motors were never used for practical purposes.

The invention of the electro-chemical battery by **Alessandro Volta** in 1799 [8] made persistent electric currents possible. **Hans Christian Ørsted** discovered in 1820 that an electric current creates a magnetic field, which can exert a force on a magnet. It only took a few weeks for **André-Marie Ampère** to develop the first formulation of the electromagnetic interaction and present the Ampère's force law, that described the production of mechanical force by the interaction of an electric current and a magnetic field [9].

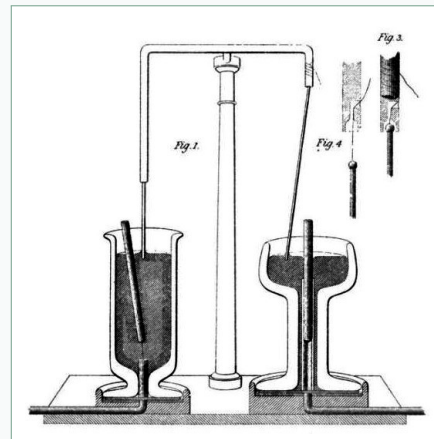


Figure 1.5
Faraday's electromagnetic experiment, 1821, the first demonstration of the conversion of electrical energy into motion [6].

Continues on next page →

← Continued from previous page

Michael Faraday gave the first demonstration of the effect with a rotary motion in 1821 [10]. A free-hanging wire was dipped into a pool of mercury, on which a PM was placed. When a current was passed through the wire, the wire rotated around the magnet, showing that the current gave rise to a close circular magnetic field around the wire [11].

1.3 Magnetic Material Properties

In the context of electromechanical energy conversion devices, the importance of magnetic materials is twofold. Through their use, it is possible to obtain **large magnetic flux densities** with relatively **low levels of magnetising force**.

As magnetic forces and energy density increase with increasing flux density, this effect plays a large role in the performance of energy-conversion devices.

In addition, magnetic materials can be used to **constrain and direct** magnetic fields in well-defined paths.

For example, in a transformer, they are used to maximise the coupling between the windings as well as to lower the excitation current¹⁸ required for transformer operation. In electric machinery, magnetic materials are used to shape the fields to obtain desired torque-production and electrical terminal characteristics.

Therefore a trained engineer can use magnetic materials to achieve specific desirable device characteristics.

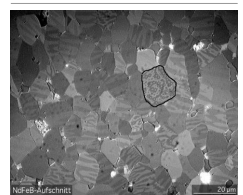
¹⁸The current or amperes required to energise the core. Even with zero load, a transformer will draw a small amount of current due to internal loss. The excitation current is made up of two components. The real component in the form of losses that are commonly referred to as no-load losses. The second form is reactive power measured in KVAR.

1.3.1 Ferromagnetic Materials

These materials are typically composed of iron and alloys of iron with cobalt, tungsten, nickel, aluminium, and other metals, are by far the most common magnetic materials. Although these materials are characterised by a wide range of properties, the basic phenomena responsible for their properties are common to them all.

Ferromagnetic materials are found to be composed of a large number of domains.¹⁹ In a non-magnetised sample of material, the domain magnetic moments are randomly oriented, and the net resulting magnetic flux in the material is zero. When an external magnetising force is applied to this material, the domain magnetic moments tend to **align with the applied magnetic field**. As a result, the domain magnetic moments add to the applied field, producing a much larger value of flux density than would exist due to the magnetising force alone. Therefore, the effective permeability (μ), equal to the ratio of the total magnetic flux density to the applied magnetic-field intensity, is large compared with the permeability of free space (μ_0). As the magnetising force is increased, this behaviour continues until all the magnetic moments are aligned with the applied field at which point they can no longer contribute to increasing the magnetic flux density, and the material is said to be **fully saturated**.

In the absence of an externally applied magnetising force, the domain magnetic moments naturally align along certain directions associated with the **crystal structure of the domain**, known as **axes of easy magnetisation**. Therefore if the applied magnetising force is reduced, the domain magnetic



¹⁹i.e., regions in which the magnetic moments of all the atoms are parallel, giving rise to a net magnetic moment for that domain [12].

Figure 1.6 shows the periodic table of elements. Elements that exhibit ferromagnetism at room temperature are highlighted in blue: Iron (Fe), Cobalt (Co), Nickel (Ni), and Manganese (Mn). The table includes element symbols, atomic numbers, and names. A legend at the bottom left identifies element categories: Alkali Metal (blue), Alkaline Earth Metal (light blue), Metal (grey), Non-metal (green), Halogen (yellow), Noble Gas (orange), and Lanthanide/Actinide (pink).

Figure 1.6

The periodic table of elements. Of all the elements that exist only a few of them have the ability to generate a magnetic field without any external need. The elements which exhibit ferromagnetism has been highlighted.

moments relax to the direction of easy magnetism nearest to that of the applied field. As a result, when the applied field is reduced to zero, although they will tend to relax towards their initial orientation, the magnetic dipole moments will no longer be totally random in their orientation; they will retain a net magnetisation component along the applied field direction. It is this effect which is responsible for the phenomenon known as magnetic hysteresis.

Due to this hysteresis effect, the relationship between **B** and **H** for a ferromagnetic material is **both non-linear and multi-valued**.

In general, material properties cannot be described analytically. They are commonly presented in graphical form as a set of empirically determined curves based on test samples.

The most common curve used to describe a magnetic material is the B-H curve or hysteresis loop which an example is shown in **Fig. 1.7**. Here, only the first quadrant is shown as it is the most important part when designing a machine.

In practice, each curve is obtained while cyclically varying the applied magnetising force between equal positive and negative values of fixed magnitude. Hysteresis causes these curves to be multi-valued. After several cycles the B-H curves form closed loops, which is shown in **Fig. 1.9**.

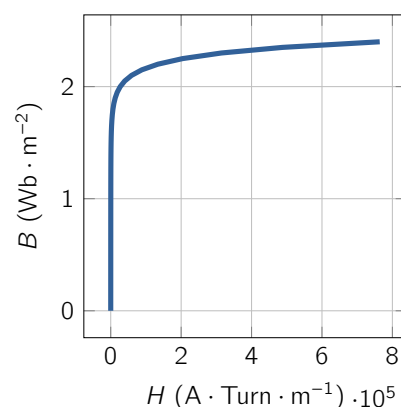


Figure 1.7

An example of a B-H curve showing only the first quadrant.

1.4 AC Excitation

In Alternating Current (AC) power systems, the wave-forms of voltage and flux **closely approximate** sinusoidal functions of time.²⁰ We will now look at the excitation characteristics and losses associated with steady-state AC operation of magnetic materials under such operating conditions. We use as our model a closed-core magnetic circuit,²¹ such as that shown in **Fig. 1.2**. The magnetic path length is l_c , and the cross-sectional area is A_c throughout the length of the core.

²⁰We say here closely as we must take into account environmental factors, such as noise and interference, which distorts the waveform.

²¹This means our circuit has no air gap.

We further assume a sinusoidal variation of the core flux $\phi(t)$, which gives:

$$\phi(t) = \phi_{\max} \sin \omega t = A_c B_{\max} \sin \omega t$$

where,

- $\phi_{\max}$ is the amplitude of core flux ϕ (Wb),
- $B_{\max}$ is the amplitude of flux density B_c (T),
- ω is the angular frequency defined as $\omega = 2\pi f$, and
- f is the frequency (Hz).

From Eq. (1.16), the voltage induced in the N -turn winding is:

$$e(t) = \omega N \phi_{\max} \cos \omega t = E_{\max} \cos \omega t$$

where

$$E_{\max} = \omega N \phi_{\max} = E_{\max} \cos \omega t$$

In steady-state AC operation, we usually are more interested in the Root-Mean Square (RMS) values of voltages and currents than their instantaneous or maximum values. In general, the RMS value of a periodic function of time, say $f(t)$, of period T is defined as:

$$F_{\text{rms}} = \sqrt{\left(\frac{1}{T} \int_0^T f^2(t) dt \right)} \quad (1.23)$$

From Eq. (1.23), the RMS value of a sine wave can be shown to be $1/\sqrt{2}$ times its peak value. Therefore, the RMS value of the induced voltage is:

$$E_{\text{rms}} = \frac{2\pi}{\sqrt{2}} f N A_c B_{\max} = \sqrt{2} \pi f N A_c B_{\max}$$

An excitation current i_ϕ , corresponding to an excitation MMF $Ni_\phi(t)$, is **required** to produce the flux $i_\phi(t)$ in the core.

Because of the non-linear magnetic properties of the core, the excitation current corresponding to a sinusoidal core flux will be **non-sinusoidal**.

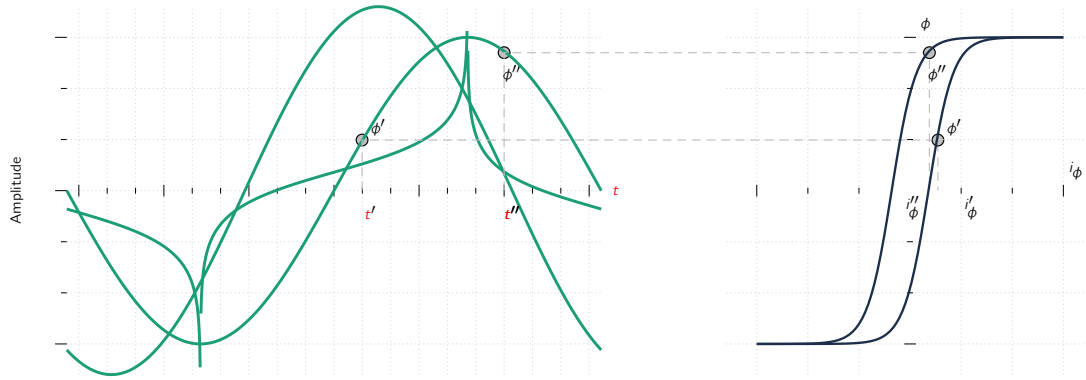


Figure 1.8

Excitation phenomena where left side shows the Voltage, flux, and exciting current and right shows the corresponding hysteresis loop. The current is not to scale.

As B_c and H_c are related to ϕ and i_ϕ by known **geometric constants**, the AC hysteresis loop of **Fig. 1.8** has been drawn in terms of $\phi = B_c A_c$ and is $i_\phi = H_c l_c / N$. The sine waves of induced voltage (e), and flux (ϕ), are shown in the left side of **Fig. 1.8**.

At any given time, the value of i_ϕ corresponding to the given value of flux can be found directly from the hysteresis loop.

As an example, at time t' the flux is ϕ' and the current is i_ϕ' , and at time t'' the corresponding values are ϕ'' and i_ϕ'' . Please observe that since the hysteresis loop is multi-valued, it is necessary to be careful to pick the rising-flux values from the **rising-flux portion** of the hysteresis loop. In the same context, the falling-flux portion of the hysteresis loop must be selected for the falling-flux values.

A second point worth mentioning is, due to the hysteresis loop “flattening out” from the **saturation effects**, the waveform of the exciting current is sharply peaked. Its RMS value $i_{\phi, \text{rms}}$ defined by Eq. (1.23), where T is the period of a cycle. It is related to the corresponding RMS value $H_{c, \text{rms}}$ of H_c by the relationship:

$$i_{\phi, \text{rms}} = \frac{l_c H_{c, \text{rms}}}{N} \quad (1.24)$$

The AC excitation characteristics of core materials are often described in terms of RMS volt-amperes ($V \cdot A$) rather than a magnetisation curve relating $\mathbf{B}$ and $\mathbf{H}$. The theory behind this representation can be explained by combining Eq. (1.24) and Eq. (1.25). Therefore, from Eq. (1.24), and Eq. (1.25) the RMS volt-amperes required to excite the core of **Fig. 1.2** to a specified flux density is equal to

$$\begin{aligned} E_{\text{rms}} i_{\phi, \text{rms}} &= \sqrt{2} \pi f N A_c B_{\text{max}} \frac{l_c H_{c, \text{rms}}}{N} \\ &= \sqrt{2} f B_{\text{max}} H_{\text{rms}} (A_c l_c) \end{aligned} \quad (1.25)$$

In Eq. (1.25), the product $A_c l_c$ can be seen to be equal to the volume of the core and hence the RMS exciting volt-amperes required to excite the core with sinusoidal can be seen to be proportional to the frequency of excitation, the core volume and the product of the peak flux density and the RMS magnetic field intensity.

For a magnetic material of mass density ρ_c , the mass of the core is $A_c l_c \rho_c$ and the exciting RMS

voltamperes per unit mass, S_a , can be expressed as

$$S_a = \frac{E_{rms} I_{\phi, rms}}{\text{mass}} = \sqrt{2} \pi f \left(\frac{B_{max} H_{rms}}{\rho_c} \right)$$

Observe that excitation of the material really depends on frequency and B_{max} . As a result, the ac excitation requirements for a magnetic material are often supplied by manufacturers in terms of RMS voltamperes per unit weight as determined by laboratory tests on closed-core samples of the material.

The exciting current supplies the MMF required to produce the core flux and the power input associated with the energy in the magnetic field in the core. Part of this energy is dissipated as losses and results in heating of the core. The rest appears as reactive power associated with energy storage in the magnetic field. This reactive power is not dissipated in the core; it is cyclically supplied and absorbed by the excitation source.

There are two (2) loss mechanisms are associated with time-varying fluxes in magnetic materials.

Eddy Current Losses The first is ohmic ($I^2 R$) heating, associated with induced currents in the core material. From Faraday's law (Eq. (1.16)) we see that time-varying magnetic fields give rise to electric fields. In magnetic materials, these electric fields result in induced currents, commonly referred to as eddy currents, which circulate in the core material and **oppose** changes in flux density in the material. To counteract the corresponding demagnetising effect, the current in the exciting winding must increase.

This causes the resultant "dynamic" B-H loop under ac operation to become somewhat "fatter" than the hysteresis loop for slowly varying conditions, and this effect increases as the excitation frequency is increased.

It is for this reason that the characteristics of electrical steels vary with frequency and therefore manufacturers typically supply characteristics over the expected operating frequency range of a particular electrical steel.

To reduce the effects of eddy currents, magnetic structures are usually built of **thin sheets of laminations** of the magnetic material.²² These laminations, which are **aligned in the direction of the field lines**, are insulated from each other by an oxide layer on their surfaces or by a thin coat of insulating enamel or varnish. This greatly reduces the magnitude of the eddy currents since the layers of insulation interrupt the current paths.

The thinner the laminations, the lower the losses.

In general, eddy-current loss tends to increase as the **square of the excitation frequency** and also as the square of the peak flux density.



²²The primary reason is to reduce the volume in which the current flows. Smaller the volume, smaller the losses [13].

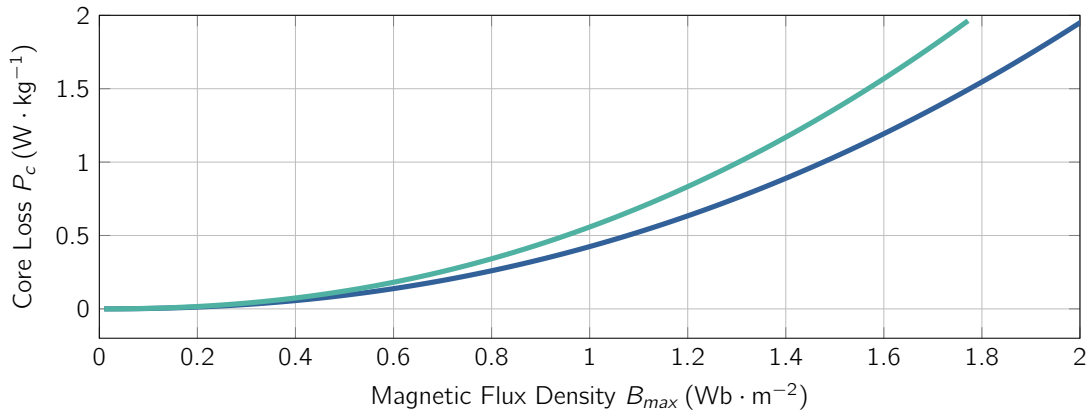


Figure 1.10

Core-loss density of M130-27S, Electric Grain steel under 50 Hz and 60 Hz. The following plot has been generated using the Steinmetz equation.

Hysteresis Losses The second loss mechanism is due to the **hysteretic nature of magnetic material**. In a magnetic circuit similar to **Fig. 1.2**, a time-varying excitation will cause the magnetic material to undergo a cyclic variation described by a hysteresis loop such as that shown in **Fig. 1.9**.

We can use Eq. (1.22) to calculate the energy input W to the magnetic core of **Fig. 1.2** as the material undergoes a single cycle:

$$W = \oint i_\phi d\lambda = \oint \left(\frac{H_c l_c}{N} \right) (A_c N dB_c) = A_c l_c \oint H_c dB_c \quad (1.26)$$

Looking at Eq. (1.26), we see that $A_c l_c$ is the volume of the core and the integral is the area of the AC hysteresis loop, and we see each time the magnetic material undergoes a cycle, there is a net energy input into the material. This energy is required to move around the magnetic dipoles in the material and is dissipated as heat in the material.

Therefore for a given flux level, the corresponding hysteresis losses are proportional to the area of the hysteresis loop and to the total volume of material.

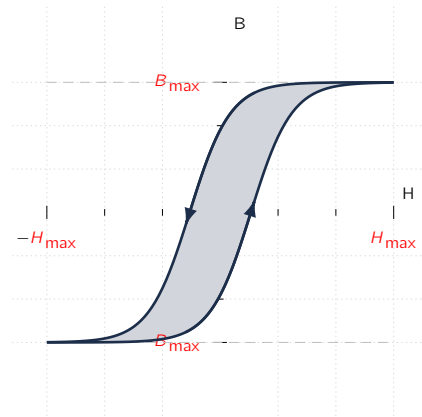


Figure 1.9

Hysteresis loop; hysteresis loss is proportional to the loop area (shaded).

Since there is an energy loss per cycle, hysteresis power loss is proportional to the frequency of the applied excitation.

In general, these losses depend on the metallurgy of the material as well as the flux density and frequency. Information on core loss is typically presented in graphical form. It is plotted in terms of watts per unit weight as a function of flux density; often a family of curves for different frequencies are given. **Fig. 1.10** shows the core loss (P_c) for M130-27S grain-oriented electrical steel at 60 Hz and 50 Hz.

Nearly all transformers and certain sections of electric machines use sheet-steel material that has **highly favourable directions of magnetisation** along which the core loss is low and the permeability

is high. This material is termed **grain-oriented steel**.

Orientation of Atoms

1.4 Information

The reason for this property lies in the atomic structure of a crystal of the silicon-iron alloy, which is a body-centred cube. Each cube has an atom at each corner as well as one in the centre of the cube. In the cube, the easiest axis of magnetisation is the cube edge; the diagonal across the cube face is more difficult, and the diagonal through the cube is the most difficult. By suitable manufacturing techniques most of the crystalline cube edges are aligned in the rolling direction to make it the favourable direction of magnetisation. The behaviour in this direction is superior in core loss and permeability to non-oriented steels in which the crystals are randomly oriented to produce a material with characteristics which are uniform in all directions. As a result, oriented steels can be operated at higher flux densities than the non-oriented grades.

Non-oriented electrical steels are used in applications where the flux does not follow a path which can be oriented with the rolling direction or where low cost is of importance. In these steels the losses are somewhat higher and the permeability is very much lower than in grain-oriented steels.

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Chapter 2

Transformers

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Before we continue on with electric drives, it is worthwhile to discuss particular aspects of the theory of magnetically-coupled circuits, with a special emphasis on the **transformer**. Although the transformer is **NOT** an energy conversion device,¹ by the strictest of the definition, it is nevertheless an **indispensable component** in many energy conversion systems and used heavily by the power generation, distribution, and transmission companies as it allows the:

¹In the end all it does is to convert one level of AC voltage to another level of AC voltage. It does **NOT** convert energy.

- electric generation at the most economical generator voltage,
- power transfer at the most economical transmission voltage, and
- power utilization at the most suitable voltage,

for the particular utilisation device. The transformer is also widely used in low-power, low-current electronic and control circuits for performing such functions as:

- matching the impedances of a source and its load for maximum power transfer,
- isolating one circuit from another, or
- isolating direct current while maintaining ac continuity between two circuits.

The transformer is one of the simpler devices made from two (**2**) or more electric circuits coupled by a **common magnetic circuit**. Its analysis involves many of the principles essential to the study of

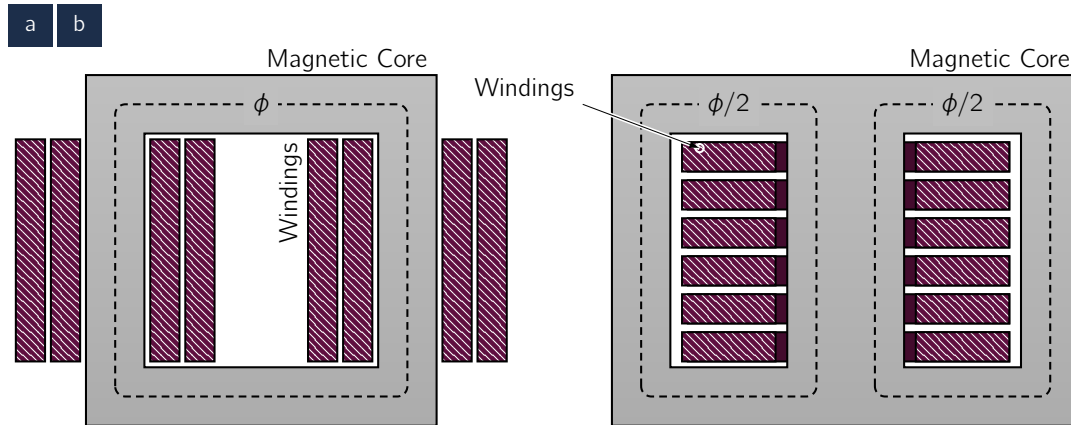


Figure 2.1
Schematic views of (a) core-type and (b) shell-type transformers.

electric drives. Therefore, our study of the transformer will serve as a bridge between the introduction to magnetic-circuit analysis we conducted previously and the more detailed study of electric drives to follow.²

2.1 The Stationary Electric Machine

Let's have a look at the machine which plays an integral role in efficient power transmission.

2.1.1 An Overview of Transformers

To give a brief definition, a transformer consists of two (2) or more windings coupled by **mutual magnetic flux**. If one of these windings, **the primary**, is connected to an AC source, alternating flux will be produced whose amplitude will depend on the primary voltage, the frequency of the applied voltage, and finally, the number of turns. The mutual flux will link the other winding, **the secondary**, and will induce a voltage in it whose value will depend on the number of secondary turns as well as the magnitude of the mutual flux and the frequency.

By properly proportioning the number of primary and secondary turns, almost **any desired voltage ratio**, or ratio of transformation, can be obtained.

The essence of transformer action requires only the existence of a **time-varying mutual flux** linking the two (2) windings. Such action can occur for two (2) windings coupled through air, but coupling between the windings can be made much more effectively using a core of iron or other ferromagnetic material, as most of the flux is then confined to a definite, high-permeability path, linking both windings.



²In 1884, Zipernowsky, Bláthy and Déri (ZBD), associated with the Ganz Works, determined open-core devices were impracticable, as they were incapable of reliably regulating voltage. In their joint 1885 patent applications for novel transformers, later called ZBD transformers, they described two designs with closed magnetic circuits where copper windings were either wound around an iron wire ring core or surrounded by an iron wire core. The two designs were the first application of the two basic transformer constructions in common use to this day, termed "core form" or "shell form".

Such a transformer is commonly called an **iron-core transformer** which makes majority of the market.³

The chapter is concerned primarily with **iron-core transformers**. As we looked at the topic previously, to reduce the losses caused by eddy currents in the core, the magnetic circuit usually consists of a **stack of thin laminations**. Two common construction types are shown in **Fig. 2.1**:

Core Shown in **Fig. 2.1a**, the windings are **wound around two legs** of a rectangular magnetic core.

Shell Shown in **Fig. 2.1b** the windings are wound **around the centre leg** of a three-legged core.

Silicon-steel laminations (0,55 mm) thick are generally used for transformers operating at frequencies below a few hundred Hz. Silicon steel has the desirable properties of low cost, low core loss, and high permeability at high flux densities.⁴ The cores of small transformers used in communication circuits at high frequencies and low energy levels are sometimes made of compressed powdered ferromagnetic alloys known as ferrites.⁵

In each of these configurations, most of the flux is confined to the core and therefore links both windings. The windings also produce additional flux, known as **leakage flux**, which links one winding without linking the other.

Although leakage flux is a small fraction of the total flux, it plays an **important role in determining the overall behavior** of the transformer which will be looked later.

In practice, leakage is reduced by subdividing the windings into sections placed as close together as possible. In the core-type construction, each winding consists of two sections, one section on each of the two legs of the core, the primary and secondary windings being **concentric coils**. In the shell-type construction, variations of the **concentric-winding** arrangement may be used, or the windings may consist of a number of thin “pancake” coils assembled in a stack with primary and secondary coils interleaved.



³ An example of an iron-core transformer.

⁴ This value ranges between 1 T to 1,5 T.



⁵ A family of iron oxide-containing magnetic ceramic materials. They are ferrimagnetic, meaning they are attracted by magnetic fields and can be magnetized to become permanent magnets. Unlike many ferromagnetic materials, most ferrites are not electrically conductive, making them useful in applications like magnetic cores for transformers to suppress eddy currents.

Distribution Transformers

2.1 Information

Provides a final voltage reduction in the electric power distribution system, stepping down the voltage used in the distribution lines to the level used by the customer.⁶ If mounted on a utility pole, they are called pole-mount transformers. When placed either at ground level or underground, distribution transformers are mounted on concrete pads and locked in steel cases, thus known as distribution tap pad-mounted transformers.

Distribution transformers typically have ratings less than 200 kVA, although some national standards allow units up to $5,000 \times 10^3$ kVA to be described as distribution transformers. Since distribution transformers are energized 24 hours a day even when they don't carry any load, **reducing iron losses is vital in their design**. They usually don't operate at full load, so they are **designed to have maximum efficiency at lower loads**. To have better efficiency, voltage regulation in these transformers is kept to a minimum. Therefore, they are **designed to have small leakage reactance**.



⁶ A single-phase distribution transformer.

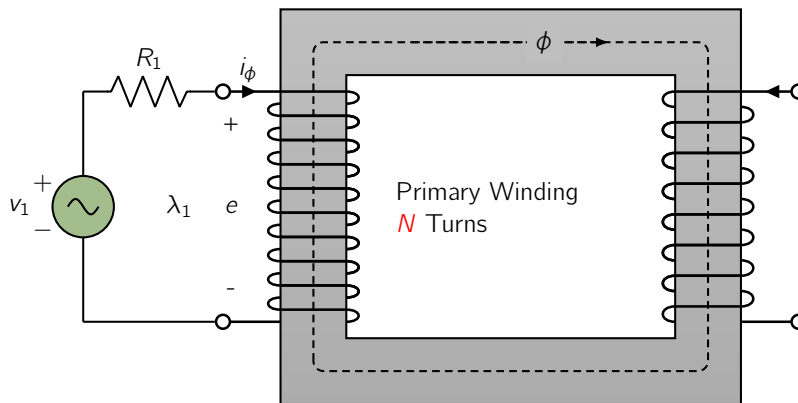


Figure 2.2
Transformer with open
secondary.

2.2 A Mathematical Model of the Transformer

To get a better understanding of how a transformer works we need to study its behavior under different conditions and determine its important parameters.

2.2.1 Transformers under No-Load Conditions

Fig. 2.2 shows in schematic form a transformer with its secondary circuit open and an alternating voltage v_1 applied to its primary terminals.

To simplify the drawings, it is common on schematic diagrams of transformers to show the primary and secondary windings as if they were on **separate legs** of the core, as in **Fig. 2.2**, even though the **in practice, the windings are actually interleaved**.

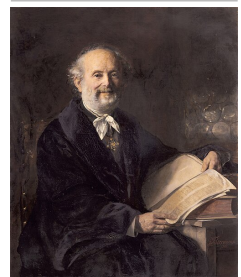
As discussed previously, a small steady-state current (i_ϕ), called the **exciting current**, flows in the primary and establishes an alternating flux in the magnetic circuit.⁷ This flux induces an Electro-motive Force (EMF) in the primary equal to:

$$e_1 = \frac{d\lambda_1}{dt} = N_1 \frac{d\phi}{dt}$$

where λ_1 is flux linkage of the primary winding (Wb),⁸ ϕ is the flux in the core linking both windings (Wb), N_1 is the number of turns in the primary winding, the voltage e_1 is in volts when ϕ is in webers. This EMF, together with the voltage drop in the primary resistance R_1 , must balance the applied voltage v_1 , therefore we can write the relation:

$$v_1 = R_1 i_\phi + e_1$$

⁷In general, the exciting current corresponds to the net ampere-turns (MMF) acting on the magnetic circuit, and it is not possible to distinguish whether it flows in the primary or secondary winding or partially in each winding.



⁸Wilhelm Eduard Weber
(1804 - 1891)

a German physicist and, together with Carl Friedrich Gauss, inventor of the first electromagnetic telegraph. The unit of magnetic flux is named after him.

Neglecting the Primary Leakage Flux

2.2 Information

At the moment we are neglecting the effects of primary leakage flux as **it would create unnecessary complications**. For all practical purposes, the flux contributes a small value to the core-flux, therefore we can neglect it. It does however play an important role in the behavior of transformers in real-life applications and taken into consideration.

If the transformer is sufficiently large, the effect of the no-load resistance can be neglected which would make the induced EMF e_1 very nearly equals the applied voltage v_1 . Furthermore, the waveforms of voltage and flux are very **nearly** sinusoidal. Therefore, if the instantaneous flux is:

$$\varphi = \phi_{\max} \sin \omega t$$

the induced voltage (e_1) is therefore:

$$e_1 = N_1 \frac{d\varphi}{dt} = N_1 \frac{d\phi_{\max} \sin \omega t}{dt} = \omega N_1 \phi_{\max} \cos \omega t$$

where $\phi_{\max}$ is the maximum value of the flux (Wb) and $\omega = 2\pi f$, with frequency being f (Hz). For the current and voltage reference directions shown in **Fig. 2.2**, the induced EMF **leads** the flux by 90° . The RMS value of the induced EMF (e_1) is:

$$E_1 = \frac{2\pi}{\sqrt{2}} f N_1 \phi_{\max} = \sqrt{2} \pi f N_1 \phi_{\max} \quad (2.1)$$

If the resistive voltage drop is assumed negligible, the counter EMF equals the applied voltage. Under these conditions, if a sinusoidal voltage is applied to a winding, a sinusoidally varying core flux must be established whose maximum value $\phi_{\max}$ satisfies the requirement that E_1 in Eq. (2.1) equal the rms value V_1 of the applied voltage. Therefore we can write:

$$\phi_{\max} = \frac{V_1}{\sqrt{2} \pi f N_1} \quad (2.2)$$

Under these conditions, the core flux is determined solely by the applied voltage (V_1), its frequency (f), and the number of turns (N) in the winding.

This relation applies not only to transformers but to **any device with a sinusoidally-alternating voltage**, as long as the resistance and leakage-inductance voltage drops are negligible.

The core flux is fixed by the applied voltage, and the required exciting current is determined by the magnetic properties of the core. The exciting current must adjust itself so as to produce the MMF required to create the flux demanded by Eq. (2.2).

Due to the non-linear magnetic properties of iron, the waveform of the exciting current differs from the waveform of the flux, which we have looked at **Fig. 1.8** in Chapter 1.

If the exciting current is analysed using *Fourier-series*, it is found to consist of a fundamental component and a series of **odd harmonics**.⁹ The fundamental component can, in turn, be resolved into two (2) components:

⁹This is caused by the function being a sine function which classify it as a Fourier odd-series.



1. one in phase with the counter EMF, and
2. the other **lagging** the counter EMF by 90° .

The in-phase component supplies the power absorbed by **hysteresis** and **eddy-current** losses in the core. It is referred to as **core-loss component of the exciting current**. When the core-loss component is subtracted from the total exciting current, the remainder is called the **magnetizing current**. It comprises a fundamental component lagging the counter EMF by 90° , together with all the harmonics. The principal harmonic is the third.¹⁰

¹⁰For typical power transformers, the third harmonic usually is about 40 % of the exciting current.

Effect of Harmonics

2.3 Information

Except in problems concerned directly with the effects of harmonic currents, the exciting-current waveform distortions can be assumed negligible, as the exciting current itself is small, especially in large transformers.

For example, the exciting current of a typical power transformer is about 1 % to 2 % of full-load current. In the end, the effects of the harmonics usually are dominated out by the sinusoidal-currents supplied to other linear elements in the circuit.

The exciting current can then be represented by an equivalent sinusoidal current which has the same RMS value and frequency and produces the same average power as the actual exciting current. Such representation is essential to the construction of a **phasor diagram**, which represents the phase relationship between the various voltages and currents in a system in vector form. Each signal is represented by a phasor whose length is proportional to the amplitude of the signal and whose angle is equal to the phase angle of that signal as measured with respect to a chosen reference signal.

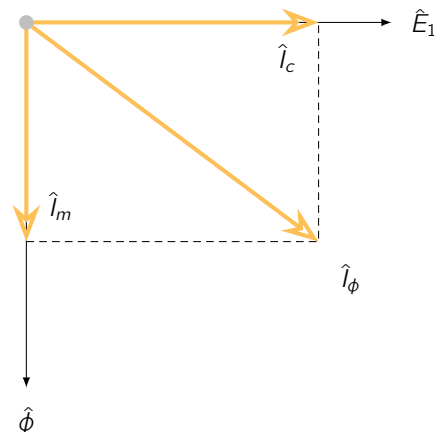


Figure 2.3
No-load phasor diagram.

In **Fig. 2.3**, the phasors $\hat{E}_1$ and $\hat{\phi}$ respectively, represent the RMS values of the induced EMF and the flux. The phasor $\hat{i}_\phi$ represents the RMS value of the equivalent sinusoidal exciting current. It lags the induced emf $\hat{E}_1$ by a phase angle θ_c . The core loss P_c , equal to the product of the in-phase components of the $\hat{E}_1$ and $\hat{i}_\phi$, is given by:

$$P_c = E_1 I_\phi \cos \theta_c$$

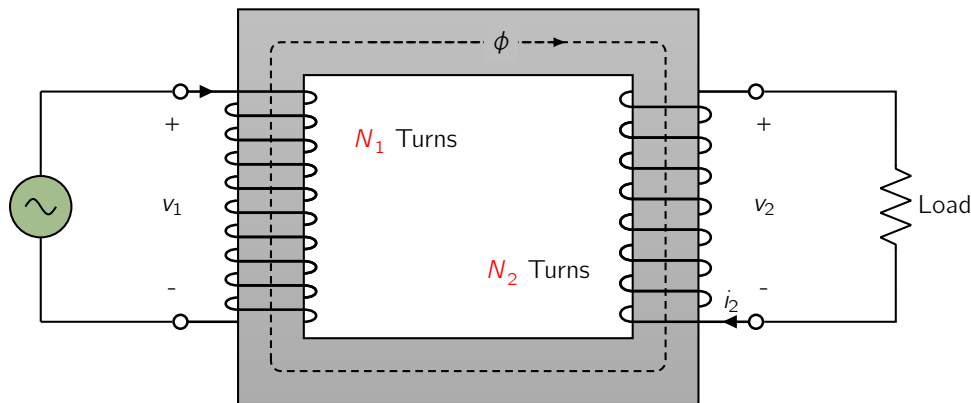


Figure 2.5

Ideal transformer and load.

Electric Arc Furnace

2.4 Information

An electric arc furnace (EAF) is a furnace that heats material by means of an electric arc.

An Industrial EAF range in size from small units of approximately one-tonne capacity, used in foundries for producing cast iron products, up to about 400-tonne units used for secondary steelmaking. Industrial EAF temperatures can reach 1,800 °C, while laboratory units can exceed 3,000 °C.

As an example, a mid-sized modern steelmaking furnace would have a transformer rated about 60,000,000 volt-amperes (60 MVA), with a secondary voltage between 400 and 900 volts and a secondary current in excess of 44,000 amperes.

In electric arc furnaces, the material inside the furnace (referred to as a charge) is directly exposed to an electric arc, and the current from the electrode terminals passes through the charge material. Arc furnaces differ from induction furnaces, which use eddy currents to heat the charge.



Figure 2.4

An electric arc furnace during operation.

2.2.2 The Effect of Secondary Current on an Ideal Transformer

As a first approximation, consider a transformer with a primary winding of N_1 turns and a secondary winding of N_2 turns, as shown schematically in **Fig. 2.5**. Please observe the secondary current is defined as positive out of the winding, therefore positive secondary current produces an MMF in the **opposite** direction from that created by positive primary current.

Let us assume the properties of this transformer are idealised under the assumption that winding resistances are negligible, and all the flux is confined to the core and links both windings,¹¹ that there are no losses in the core, and that the permeability of the core is so high that only a negligible exciting MMF is required to establish the flux.

¹¹This means the leakage flux is assumed negligible.

These properties are **closely approached but never actually attained** in practical transformers.

A hypothetical transformer having these properties is often called an **ideal transformer**. Under the above assumptions, when a time-varying voltage v_1 is given on the primary terminals, a core flux φ must be established such that the counter emf e_1 equals the impressed voltage. Therefore:

$$v_1 = e_1 = N_1 \frac{d\varphi}{dt} \quad (2.3)$$

The core flux also links the secondary and produces an induced EMF e_2 , and an equal secondary terminal voltage v_2 , given by

$$v_2 = e_2 = N_2 \frac{d\varphi}{dt} \quad (2.4)$$

From the ratio of Eq. (2.3) and Eq. (2.4):

$$\frac{v_1}{v_2} = \frac{N_1}{N_2} \quad (2.5)$$

An ideal transformer transforms voltages in the direct ratio of the turns in its windings.

Now let's have a load and connect it to the secondary. A current i_2 and an MMF $N_2 i_2$ are then present in the secondary. Given that the core permeability is assumed very large and since the impressed primary voltage sets the core flux as specified by Eq. (2.3), the core flux is unchanged by the presence of a load on the secondary, and hence the net exciting MMF acting on the core¹² will **NOT** change and hence will remain negligible. Therefore

¹²which is equal to $N_1 i_1 - N_2 i_2$.

$$N_1 i_1 - N_2 i_2 = 0 \quad (2.6)$$

From Eq. (2.6) we see that a compensating primary MMF must result to cancel that of the secondary. We can write it as:

$$N_1 i_1 = N_2 i_2 \quad (2.7)$$

Therefore we see that the requirement that the net MMF **remain unchanged** is the means by which the primary "knows" of the presence of load current in the secondary as any change in MMF flowing in the secondary as the result of a load must be accompanied by a corresponding change in the primary MMF.

For the reference directions shown in **Fig. 2.5** the mmf's of i_1 and i_2 are in opposite directions and therefore compensate.

The net MMF acting on the core therefore is zero, in accordance with the assumption that the exciting current of an ideal transformer is zero. From Eq. (2.7):

$$\frac{i_1}{i_2} = \frac{N_2}{N_1} \quad (2.8)$$

Therefore an ideal transformer transforms currents in the **inverse ratio** of the turns in its windings. Also notice from Eq. (2.5) and Eq. (2.8) that:

$$v_1 i_1 = v_2 i_2$$



This means, the instantaneous power input to the primary equals the instantaneous power output from the secondary, a necessary condition because all dissipative and energy storage mechanisms in the transformer have been neglected.

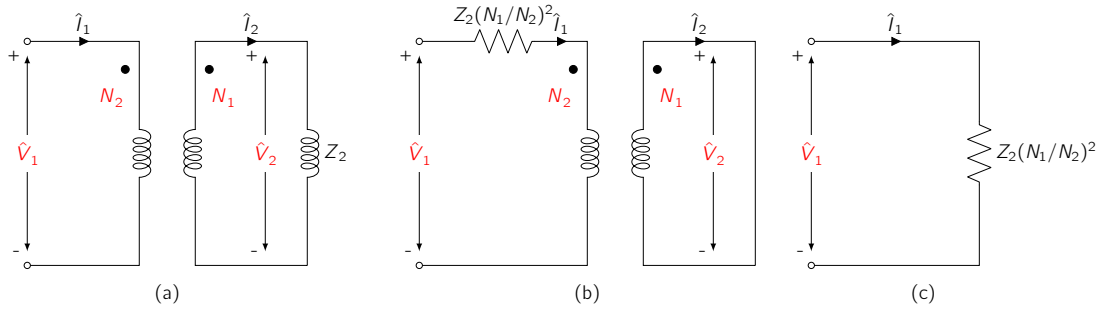


Figure 2.6: Three circuits which are identical at the primary terminals when the transformer is ideal.

An additional property of the ideal transformer can be seen by considering the case of a sinusoidal applied voltage and an impedance load. The circuit is shown in simplified form in **Fig. 2.6a**, in which the dotmarked terminals of the transformer correspond to the similarly marked terminals in **Fig. 2.5**.

The dot markings indicate terminals of corresponding polarity.

For example, if one follows through the primary and secondary windings of **Fig. 2.5**, beginning at their dot-marked terminals, one will find that both windings encircle the core in the same direction with respect to the flux.

Therefore, if one compares the voltages of the two windings, the voltages from a dot-marked to an unmarked terminal will be of the same instantaneous polarity for primary and secondary.

In other words, the voltages $\hat{V}_1$ and $\hat{V}_2$ in **Fig. 2.6a** are in phase. Also currents $\hat{I}_1$ and $\hat{I}_2$ are in phase as seen from Eq. (2.7).

The polarity of $\hat{I}_1$ is defined as into the dotted terminal and the polarity of $\hat{I}_2$ is defined as out of the dotted terminal.

We next investigate the impedance transformation properties of the ideal transformer. In phasor form, Eq. (2.5) and Eq. (2.8) can be expressed as

$$\begin{aligned} \hat{V}_1 &= \frac{N_1}{N_2} \hat{V}_2 \quad \text{and} \quad \hat{V}_2 = \frac{N_2}{N_1} \hat{V}_1 \\ \hat{I}_1 &= \frac{N_2}{N_1} \hat{I}_2 \quad \text{and} \quad \hat{I}_2 = \frac{N_1}{N_2} \hat{I}_1 \end{aligned} \quad (2.9)$$

From these equations we can derive the following relation:

$$\frac{\hat{V}_1}{\hat{I}_1} = \left(\frac{N_1}{N_2} \right)^2 \frac{\hat{V}_2}{\hat{I}_2}$$

Noting that the load impedance Z_2 is related to the secondary voltages and currents

$$Z_2 = \frac{\hat{V}_2}{\hat{I}_2}$$

where Z_2 is the **complex impedance of the load**. Consequently, as far as its effect is concerned, an impedance Z_2 in the secondary circuit can be replaced by an equivalent impedance Z_1 in the primary circuit, provided the following condition is assumed true:

$$Z_1 = \left(\frac{N_1}{N_2} \right)^2 Z_2$$

Therefore, the three circuits of **Fig. 2.6** are indistinguishable as far as their performance viewed from terminals ab is concerned. Transferring an impedance from one side of a transformer to the other in this fashion is called **referring the impedance** to the other side.

Impedances transform as the **square of the turns ratio**.

In a similar manner, voltages and currents can be **referred** to one side or the other by using Eq. (2.9) to evaluate the equivalent voltage and current on that side.

2.2.3 Reactances and Types of Equivalent Circuits

The departures of an actual transformer from those of an ideal transformer must be included to a greater or lesser degree in most analyses of transformer performance. A more complete model must take into account the effects of **winding resistances**, **leakage fluxes**, and **finite exciting current** due to the finite¹³ permeability of the core.

¹³and indeed non-linear

High Frequency Effects

2.5 Information

In some cases, the capacitances of the windings also have important effects, notably in problems involving transformer behavior at frequencies above the audio range or during rapidly changing transient conditions such as those encountered in power system transformers as a result of voltage surges caused by lightning or switching transients.

The analysis of these high-frequency problems is beyond the scope of the lecture. In addition, to keep things simple, the capacitances of the windings will be neglected.

Two methods of analysis by which departures from the ideal can be taken into account:

Physical Reasoning

Developing an equivalent-circuit model of the transformer which all physical effects are represented as circuit elements.

Magnetic Circuit

Studying the behavior of the transformer using magnetically coupled circuits which tries to develop the model from a magnetic material perspective.

Both methods are in everyday use, and both have very close parallels in the theories of rotating

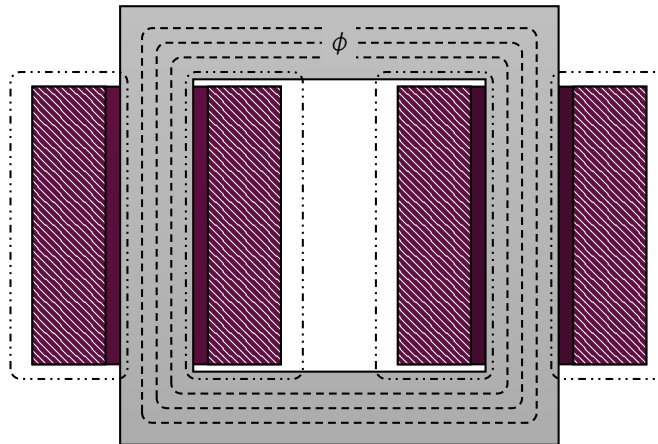


Figure 2.7

Schematic view of mutual and leakage fluxes in a transformer. The paths which are drawn densely dashed are the resultant magnetic flux paths, and paths drawn using dashed dot are leakage path on the primary (LHS) and secondary (RHS) respectively.

machines. However, as it offers an excellent example of the thought process involved in translating physical concepts to a quantitative theory, the equivalent circuit technique is presented here.

Deriving the Circuit Model

To begin the development of a transformer equivalent circuit, we first consider the primary winding. The total flux linking the primary winding can be divided into two (2) components:

1. The resultant mutual flux, confined essentially to the iron core and produced by the combined effect of the primary and secondary currents, and
2. The primary leakage flux, which links only the primary.

These components are identified in the schematic transformer shown in **Fig. 2.7**, where for simplicity the primary and secondary windings are shown on opposite legs of the core.¹⁴

The leakage flux induces voltage in the primary winding which adds to the mutual flux. Given the leakage path is largely in air, this flux and the voltage induced by it vary **linearly** with primary current i_1 . It can therefore be represented by a **primary leakage inductance** L_1 .¹⁵ The corresponding **primary leakage reactance** X_{l_1} is found as

$$X_{l_1} = 2\pi f L_1$$

Of course, it is also worth mentioning, there will be a voltage drop in the primary resistance R_1 . We now see the primary terminal voltage $\hat{V}_1$ consists of three components:

- the $\hat{I}_1 R_1$ drop in the primary resistance,
- the $j \hat{I}_1 X_{l_1}$ drop arising from primary leakage flux, and

¹⁴In an actual transformer with interleaved windings, the details of the flux distribution are more complicated, but the essential features remain the same.

¹⁵Which is equal to the leakage-flux linkages with the primary per unit of primary current.



- the EMF $\hat{E}_1$ induced in the primary by the resultant mutual flux.

Fig. 2.10a shows an equivalent circuit for the primary winding which includes each of these voltages.

The resultant mutual flux links both the primary and secondary windings and is created by their combined MMF's. It is convenient to treat these mmf's by considering the primary current must meet two requirements of the magnetic circuit:

1. It must not only produce the MMF required to produce the resultant mutual flux,
2. It must also counteract the effect of the secondary MMF which acts to demagnetize the core.

An alternative viewpoint is that the primary current must not only magnetize the core, it must also supply current to the load connected to the secondary. According to this picture, it is convenient to resolve the primary current into two (2) components:

1. an exciting component, and
2. a load component.

The exciting component $\hat{I}_\phi$ is defined as the additional primary current required to produce the resultant mutual flux. It is a nonsinusoidal current of the nature described in subsection 2.2.1.¹⁶ The load component $\hat{I}'_2$ is defined as the component current in the primary which would exactly counteract the mmf of secondary current $\hat{I}_2$. Given it is the exciting component which produces the core flux, the net mmf must equal $N_1 \hat{I}_\phi$ and thus we see that

$$N_1 \hat{I}_\phi = N_1 \hat{I}_1 - N_2 \hat{I}_2 = N_1 (\hat{I}_\phi + \hat{I}'_2) - N_2 \hat{I}_2 \quad (2.10)$$

and from Eq. (2.10) we see that

$$\hat{I}'_2 = \frac{N_2}{N_1} \hat{I}_2 \quad (2.11)$$

From Eq. (2.11), we see that the load component of the primary current equals the secondary current referred to the primary exactly as in an ideal transformer.

The exciting current can be treated as an equivalent sinusoidal current $\hat{I}_\phi$, and can be resolved into a core-loss component $\hat{I}_c$ in phase with the emf $\hat{E}_1$ and a magnetizing component $\hat{I}_m$ lagging $\hat{E}_1$ by 90° . In the equivalent circuit (Fig. 2.10b) the equivalent sinusoidal exciting current is accounted for by means of a shunt branch connected across $\hat{E}_1$, comprising a coreloss resistance R_c in parallel with a magnetizing inductance L_m whose reactance, known as the magnetizing reactance, is given by

$$X_m = 2\pi f L_m$$

In the equivalent circuit of (Fig. 2.10b) the power E_1^2/R_c accounts for the core loss due to the resultant mutual flux. R_c is referred to as the magnetizing resistance or **core loss** resistance and together with X_m forms the excitation branch of the equivalent circuit, and we will refer to the parallel combination of R_c and X_m as the exciting impedance Z_ϕ .

When R_c is assumed constant, the core loss is thereby assumed to vary as E_1^2 or as $\phi_{\max}^2 f^2$, where

¹⁶In fact, the exciting current corresponds to the net MMF acting on the transformer core and cannot, in general, be considered to flow in the primary alone. However, for the purposes of this discussion, this distinction is not significant.

$\phi_{\max}$ is the maximum value of the resultant mutual flux. Strictly speaking, the magnetizing reactance (X_m) varies with the saturation of the iron.

When X_m is assumed constant, the magnetizing current is thereby assumed to be **independent** of frequency and directly proportional to the resultant mutual flux. Both R_c and X_m are usually determined at rated voltage and frequency; they are then assumed to remain constant for the small departures from rated values associated with normal operation.

We will next add to our equivalent circuit a representation of the secondary winding. We begin by recognizing that the resultant mutual flux (Φ) induces an EMF E_2 in the secondary, and given this flux links both windings, the induced-emf ratio must equal the winding turns ratio, i.e.,

$$\frac{\hat{E}_1}{\hat{E}_2} = \frac{N_1}{N_2}$$

just as in an ideal transformer. This voltage transformation and the current transformation of Eq. (2.11) can be accounted for by introducing an ideal transformer in the equivalent circuit, as in Fig. 2.10c. Just as is the case for the primary winding, the EMF ($\hat{E}_2$) is not the secondary terminal voltage, however, because of the secondary resistance R_2 and because the secondary current ($\hat{I}_2$) creates secondary leakage flux (see Fig. 2.9). The secondary terminal voltage ($\hat{V}_2$) differs from the induced voltage $\hat{E}_2$ by the voltage drops due to secondary resistance (R_2) and secondary leakage reactance (X_{l2}),¹⁷ as in the portion of the complete transformer equivalent circuit (Fig. 2.10c) to the right of E_2 .

¹⁷This corresponds to the secondary leakage inductance L_{l2} .

From the equivalent circuit of Fig. 2.10, the actual transformer therefore can be seen to be equivalent to an ideal transformer plus external impedances. By referring all quantities to the primary or secondary, the ideal transformer in Fig. 2.10c can be moved out to the right or left, respectively, of the equivalent circuit. This is almost invariably done, and the equivalent circuit is usually drawn as in Fig. 2.10d, with the ideal transformer not shown and all voltages, currents, and impedances referred to either the primary or secondary winding. Specifically, for Fig. 2.10d,

$$\begin{aligned} X'_{l2} &= \left(\frac{N_1}{N_2}\right)^2 X_{l2} & \text{and} & & V'_2 &= \frac{N_1}{N_2} V_2 \\ R'_2 &= \left(\frac{N_1}{N_2}\right)^2 R_2 \end{aligned}$$

The circuit of Fig. 2.10d is called the equivalent-T circuit for a transformer. In Fig. 2.10d, in which the secondary quantities are referred to the primary. Here, the referred secondary values are indicated with primes, for example, X'_1 , and R'_2 , to distinguish them from the actual values of Fig. 2.10c.

Going Forward with Referred Values

2.6 Information

In the discussion that follows **we almost always deal with referred values**, and the primes will be omitted.

One must simply keep in mind the side of the transformers to which all quantities have been referred.

2.3 Understanding Transformer Behaviour

In analysis involving the transformer as a circuit element, it is useful to adopt one of several approximate forms of the equivalent circuit¹⁸ of Fig. 2.10 rather than the full circuit. The approximations chosen in a particular case depend largely on physical reasoning based on orders of magnitude of the neglected quantities.

The approximate equivalent circuits commonly used for constant-frequency power transformer analyses are summarized for comparison in Fig. 2.12. All quantities in these circuits are referred to either the primary or the secondary, and the ideal transformer is **NOT** shown.

Calculations can often be greatly simplified by moving the shunt branch representing the exciting current out from the middle of the T-circuit to either the primary or the secondary terminals, as in Fig. 2.12a and b.

These forms of the equivalent circuit are referred to as cantilever circuits.

The series branch is the combined resistance and leakage reactance of the primary and secondary, referred to the same side. This impedance is sometimes called the equivalent series impedance (Z_{eq}) and its components are:

- the equivalent series resistance (R_{eq}), and
- equivalent series reactance (X_{eq}),

are shown in Fig. 2.12a and b.

As compared to the *equivalent-T* circuit of Fig. 2.10d, the cantilever circuit introduces additional error in that it **neglects the voltage drop** in the primary or secondary leakage impedance caused by the exciting current. As the impedance of the exciting branch is typically quite large in large power transformers, the corresponding exciting current is quite small.

This error becomes **insignificant in most situations involving large transformers**.

Further analytical simplification can be achieved by neglecting the exciting current **entirely**, as in Fig. 2.12c, in which the transformer is represented as an equivalent series impedance. If the transformer is large,¹⁹ the equivalent resistance (R_{eq}) becomes measurably insignificant compared to the equivalent reactance (X_{eq}) and therefore can generally be neglected, giving the equivalent circuit of Fig. 2.12d.

For most power system problems and analysis the circuits shown in Fig. 2.12c and d are sufficiently accurate.

As a final note, in situations where the currents and voltages are determined **almost wholly by the**

¹⁸ A theoretical circuit that retains all of the electrical characteristics of a given circuit. Often, an equivalent circuit is sought that simplifies calculation, and more broadly, that is a simplest form of a more complex circuit in order to aid analysis. But then again, we've looked at this the previous chapter so I hope this aforementioned explanation was superfluous.

¹⁹ In this context we would be talking about several hundred kilovoltamperes or more

circuits external to the transformer or when a high degree of accuracy is **NOT** required, the entire transformer impedance can be neglected and the transformer considered to be ideal, as mentioned previously.

The circuits of Fig. 2.12 have the additional advantage being the total equivalent resistance (R_{eq}) and equivalent reactance (X_{eq}) can be found from a very simple test in which one terminal is short-circuited. Whereas, the process of determining the individual leakage reactances (X_{11} and X_{12}) and a complete set of parameters for the *equivalent-T* circuit of Fig. 2.10c is, as can be expected, more difficult.

Without some prior knowledge of the turns ratio,²⁰ it is not possible to make a set of measurements which uniquely determine the turns ratio, the **magnetizing inductance**, and the **individual leakage impedances**.

²⁰based for example upon knowledge of the internal construction of the transformer.

It can be shown that, simply from terminal measurements, neither the turns ratio, the magnetizing reactance, or the leakage reactances are unique characteristics of a transformer equivalent circuit.

As an example, the turns ratio can be chosen arbitrarily and for each choice of turns ratio, there will be a corresponding set of values for the leakage and magnetizing reactances which matches the measured characteristic.

Each of the resultant equivalent circuits will have the same electrical terminal characteristics, a fact which has the fortunate consequence that any self-consistent set of empirically determined parameters will adequately represent the transformer.

There are two (2) simple tests employed to determine the parameters of the equivalent circuits of Fig. 2.10 and 2.12. These consist of measuring the input voltage, current, and power to the primary:

- first with the secondary short-circuited, and
- then with the secondary open-circuited,

which are looked in detail below.

2.3.1 Short-Circuit Test

The short-circuit test can be used to find the equivalent series impedance:

$$Z_{eq} = R_{eq} + jX_{eq}.$$

Although the choice of winding to short-circuit is arbitrary, for the sake of this discussion we shall consider the short circuit to be applied to the transformer secondary and voltage applied to primary.

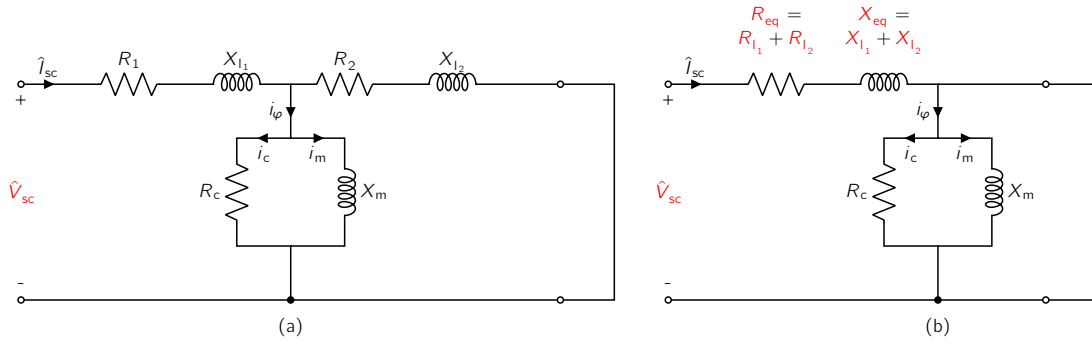


Figure 2.8: Equivalent circuit with short-circuited secondary. (a) Complete equivalent circuit. (b) Cantilever equivalent circuit with the exciting branch at the transformer secondary.

Practical Testing of Short-Circuit

2.7 Information

For convenience, the **high-voltage side** is usually taken as the primary in this test due to the **equivalent series impedance is relatively small**, typically an applied primary voltage on the order of 10 to 15 percent or less of the rated value will result in rated current.

Figure 2.15a shows the equivalent circuit with transformer secondary impedance referred to the primary side and a short circuit applied to the secondary. The short-circuit impedance (Z_{sc}) looking into the primary under these conditions is:

$$Z_{sc} = R_1 + jX_{l1} + \frac{Z_\phi(R_2 + jX_{l2})}{Z_\phi + R_2 + jX_{l2}}$$

Because the impedance of the exciting branch (Z_ϕ) is much larger than that of the secondary leakage impedance,²¹ the short-circuit impedance can be approximated as

$$Z_{sc} \approx R_1 + jX_{l1} + R_2 + jX_{l2} = R_{eq} + jX_{eq}$$

²¹which will be true unless the core is heavily saturated by excessive voltage applied to the primary; certainly not the case here

The approximation made here is equivalent to the approximation made in reducing the equivalent-T circuit to the cantilever equivalent.

This can be seen from Fig. 2.15b. The impedance seen at the input of this equivalent circuit is clearly:

$$Z_{sc} = Z_{eq} = R_{eq} + jX_{eq}$$

given the exciting branch is directly shorted out by the short on the secondary.

Typically the instrumentation used for this test will measure the RMS magnitude of the applied voltage (V_{sc}), the short-circuit current (I_{sc}), and the power (P_{sc}). Based upon these three (3)

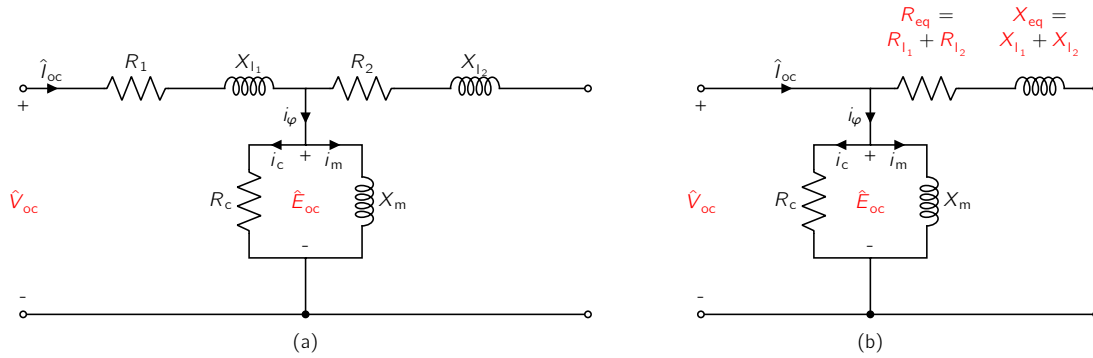


Figure 2.9: Equivalent circuit with open-circuited secondary. (a) Complete equivalent circuit. (b) Cantilever equivalent circuit with the exciting branch at the transformer primary.

measurements, the equivalent resistance and reactance²² can be found from:

²²Both being referred to the primary.

$$\begin{aligned}
 |Z_{eq}| &= |Z_{sc}| = \frac{V_{sc}}{I_{sc}} \\
 R_{eq} &= R_{sc} = \frac{P_{sc}}{I_{sc}^2} \\
 X_{eq} &= X_{sc} = \sqrt{|Z_{sc}|^2 - R_{sc}^2}
 \end{aligned} \tag{2.12}$$

The equivalent impedance can, of course, be referred from one side to the other in the usual manner. On the rare occasions when the equivalent-T circuit in Fig. 2.10d must be resorted to, approximate values of the individual primary and secondary resistances and leakage reactances can be obtained by assuming:

$$R_1 = R_2 = 0.5R_{eq} \quad \text{and} \quad X_{l1} = X_{l2} = 0.5X_{eq}$$

when all impedances are referred to the same side.

Strictly speaking, of course, it is possible to measure R_1 and R_2 directly by a Direct Current (DC) resistance measurement on each winding.²³

²³and then referring one or the other to the other side of the idea transformer

However, **no such simple test exists** for the leakage reactances X_{l1} and X_{l2} .

2.3.2 Open-Circuit Test

The open-circuit test is performed with the **secondary open-circuited** and rated voltage impressed on the primary. Under this condition an exciting current of a few percent of full-load current²⁴ is obtained. Rated voltage is chosen to ensure that the magnetizing reactance will be operating at a flux level close to that which will exist under normal operating conditions. If the transformer is to be used at other than its rated voltage, the test should be done at that voltage.

²⁴less on large transformers and more on smaller ones.

For convenience, the **low-voltage side** is usually taken as the primary in this test.



If the primary in this test is chosen to be the opposite winding from that of the short-circuit test, one must of course be careful to refer the various measured impedances to the same side of the transformer in order to obtain a self-consistent set of parameter values. Figure 2.16a shows the equivalent circuit with the transformer secondary impedance referred to the primary side and the secondary open-circuited. The open-circuit impedance Z_{oc} looking into the primary under these conditions is

$$Z_{oc} = R_1 + jX_{l1} + Z_\phi = R_1 + jX_{l1} + \frac{R_c(jX_m)}{R_c + jX_m} \quad (2.13)$$

Because the impedance of the exciting branch is quite large, the voltage drop in the primary leakage impedance caused by the exciting current is typically negligible, and the primary impressed voltage $\hat{V}_{oc}$ very nearly equals the emf $\hat{E}_{oc}$ induced by the resultant core flux.

Similarly, the primary $I_{oc}^2 R_1$ loss caused by the exciting current is negligible, so the power input P_{oc} very nearly equals the core loss E_{oc}^2/R_c . As a result, it is common to ignore the primary leakage impedance and to approximate the open-circuit impedance as being equal to the magnetizing impedance:

$$Z_{oc} \approx Z_\phi = \frac{R_c(jX_m)}{R_c + jX_m}$$

The approximation made here is equivalent to the approximation made in reducing the equivalent-T circuit to the cantilever equivalent circuit of Fig. 2.16b

The impedance seen at the input of this equivalent circuit is clearly Z_ϕ since no current will flow in the open-circuited secondary. As with the short-circuit test, typically the instrumentation used for this test will measure the rms magnitude of the applied voltage, V_{oc} , the open-circuit current I_{oc} and the power P_{oc} . Neglecting the primary leakage impedance and based upon these three measurements, the magnetizing resistance and reactance²⁵ can be found from

²⁵referred to the primary.

$$R_c = \frac{V_{oc}^2}{P_{oc}} \quad \text{and} \quad X_m = \frac{1}{\sqrt{(1/|Z_\phi|)^2 - (1/R_c)^2}}$$

$$|Z_\phi| = \frac{V_{oc}}{I_{oc}}$$

The values obtained are, of course, referred to the side used as the primary in this test. The open-circuit test can be used to obtain the core loss for efficiency computations and to check the magnitude of the exciting current. Sometimes the voltage at the terminals of the open-circuited secondary is measured as a check on the turns ratio.

Measure from the Other Side

2.8 Information

If desired, a slightly more accurate calculation of X_m and R_c by retaining the measurements of R_1 and X_{l1} obtained from the short-circuit test,²⁶ and basing the derivation on Eq. (2.13). However, such additional effort is rarely necessary for the purposes of engineering accuracy.

²⁶referred to the proper side of the transformer.

2.4 Transformers in the Wild

We've now developed the necessary mathematical and conceptual understanding required to appreciate the utility of transformers and as we can imagine, there are a wide variety of transformer types available which are designed for specific purposes which this section aims to cover.

2.4.1 Auto and Multiwinding Transformers

The core concept we have developed up to now were for two-winding transformers only. However, these can also be applied to transformers with other winding configurations such as **auto-transformers** and **multi-winding** transformers which will be the focus here.

AutoTransformers

In **Fig. 2.10a**, a two-winding transformer is shown with N_1 and N_2 turns on the primary and secondary windings respectively. The same transformation effect on voltages, currents, and impedances can also be obtained when these windings are connected as shown in **Fig. 2.10b**.

In **Fig. 2.10b**, winding bc is common to both the primary and secondary circuits.

This type of transformer is called an **autotransformer**. From construction point of view, it is little more than a normal transformer connected in a special way.

However, an important difference is the windings of the two-winding transformer are **electrically isolated** whereas those of the autotransformer are connected directly together. In addition, in the autotransformer connection, winding ab must be provided with **extra insulation** as it must be insulated against the full maximum voltage of the autotransformer.

They have lower leakage reactances, lower losses, and smaller exciting current and cost less than two-winding transformers when the voltage ratio does not differ too greatly from 1 : 1.

The following example illustrates the benefits of an autotransformer for those situations where electrical isolation between the primary and secondary windings is **NOT** an important consideration.

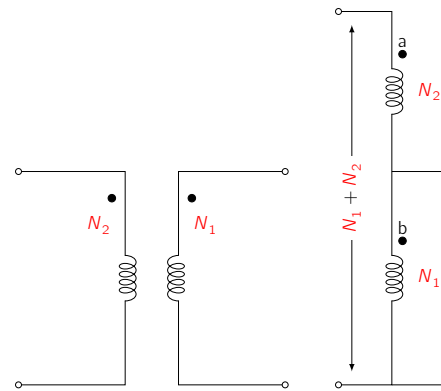


Figure 2.10: The schematics of an autotransformer



Figure 2.11: An example of an auto transformer.

When a transformer is connected as an autotransformer as shown in **Fig. 2.10**, the rated voltages of the autotransformer can be expressed in terms of those of the two-winding transformer as:

$$\begin{aligned} \text{Low-Voltage} \quad V_{L_{\text{rated}}} &= V_{1_{\text{rated}}} \\ \text{High-Voltage} \quad V_{H_{\text{rated}}} &= V_{1_{\text{rated}}} + V_{2_{\text{rated}}} = \left(\frac{N_1 + N_2}{N_1} \right) V_{L_{\text{rated}}} \end{aligned}$$

The effective turns ratio of the autotransformer is $(N_1 + N_2)/N_1$ and the power rating of the autotransformer is equal to $(N_1 + N_2)/N_2$ times that of the two-winding transformer, although the actual power processed by the transformer will **NOT** increase over that of the standard two-winding connection.

Multiwinding Transformers

Transformers having three (3) or more windings, known as **multi-winding** or **multi-circuit** transformers, are often used to interconnect three or more circuits which may have different voltages.²⁷

Transformers having a primary and multiple secondaries are frequently found in multiple-output DC power supplies for electronic applications. Distribution transformers used to supply power for domestic purposes usually have two 120 V secondaries connected in series. Circuits for lighting and low-power applications are connected across each of the 120 V windings, while electric ranges, domestic hot-water heaters, clothes-dryers, and other high-power loads are supplied with 240 V power from the series-connected secondaries.

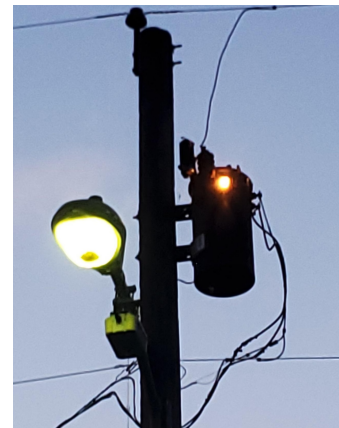


Figure 2.12

An example of a distribution transformer use for street lighting purposes.²⁷ For these applications a multiwinding transformer costs less and is more efficient than an equivalent number of two-winding transformers.

Distribution Systems In a similar fashion, a large distribution system can be supplied through a three-phase bank of multiwinding transformers from two or more transmission systems having different voltages. In addition, the three-phase transformer banks used to interconnect two transmission systems of different voltages often have a third, or tertiary, set of windings to provide voltage for auxiliary power purposes in substations or to supply a local distribution system.

Static capacitors or synchronous condensers can be connected to the tertiary windings for power factor correction or voltage regulation.²⁸ Some of the issues arising in the use of multiwinding transformers are associated with the effects of **leakage impedances** on voltage regulation, short-circuit currents, and division of load among circuits. These problems can be solved by an equivalent circuit technique mentioned previously.

²⁸Sometimes Δ -connected tertiary windings are put on three-phase banks to provide a low-impedance path for third harmonic components of the exciting current to reduce third-harmonic components of the neutral voltage.

Utility Comes at a Cost of Complexity

2.9 Information

Unfortunately equivalent circuits of multi-winding transformers are more complicated as they must take into account the leakage impedances associated with each pair of windings. Typically, in these equivalent circuits, all quantities are referred to a common base by use of the appropriate turns ratios as referring factors.

The **exciting current** is generally neglected in modelling for simplicity.

2.4.2 Poly-Phase Transformers

One of the more practical applications of transformers are, of course in the use of the grid. Given most supplied energy across vast distances are poly-phase;²⁹ we require the use of poly-phase transformers to **step-up** or **step-down** voltages as required.

²⁹Disregarding specialty cases, this is mostly three.

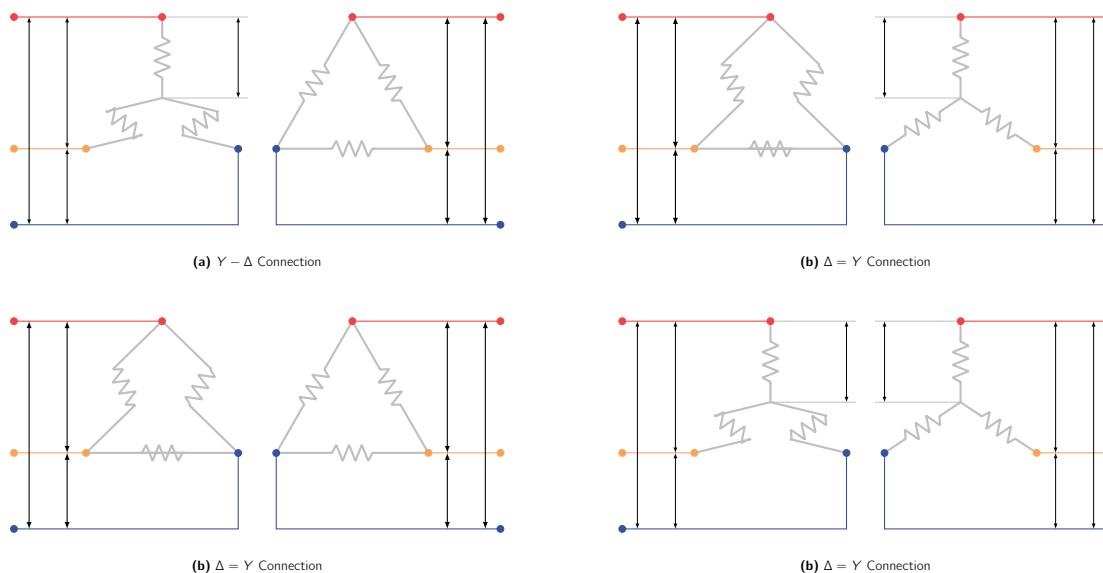


Figure 2.13: Common three-phase transformer connections; the transformer windings are indicated by the heavy lines.

Three single-phase transformers can be connected to form a three-phase transformer bank in any of the four (4) ways shown in Fig. 2.13. In all four parts of the figure, the windings at the left are the **primaries**, those at the right are the **secondaries**, and any primary winding in one transformer corresponds to the secondary winding drawn parallel to it.

Also shown are the voltages and currents resulting from balanced impressed primary line-to-line voltages (V_{LL}) and line currents (I_{LL}) when the ratio of primary-to-secondary turns is $N_1/N_2 = a$ and ideal transformer behaviour are assumed.

The rated voltages and currents at the primary and secondary of the three-phase transformer bank depends on the connection used but the rated kVA of the three-phase bank is three times that of the individual single-phase transformers, regardless of the connection.

Connection Types

As can be seen the ability to connect transformers as either Y or Δ allows us to do certain combinations by changing how the primary or the secondary is connected. Let's have a look at them and see their advantages and disadvantages.

Star-Delta Connection

Commonly used in stepping down from a high voltage to a medium or low voltage. One reason is for this is the presence of a neutral provided for grounding on the high-voltage side, a procedure which can be shown to be desirable in many cases.

Delta-Star Connection

Commonly used for stepping up to a high voltage.

Delta-Delta Connection

It has the advantage that one transformer can be removed for repair or disconnected for maintenance while the remaining two continue to function as a three-phase bank with the rating reduced to 58 percent of that of the original bank.

This is known as the **open-delta**, or V, connection.

Star-Star Connection

Seldom used because of difficulties with exciting-current phenomena.³⁰

Instead of three single-phase transformers, a three-phase bank may consist of one three-phase transformer having all six (6) windings on a **common multi-legged** core and contained in a single tank. Advantages of three-phase transformers over connections of three single-phase transformers are that they cost less, weigh less, require less floor space, and have somewhat higher efficiency. A photograph of the internal parts of a large three-phase transformer is shown in **Fig. 2.14**.

Calculations involving three-phase transformer banks **under balanced conditions** can be made by dealing with only one of the transformers or phases and recognizing that conditions are the same in the other two phases except for the phase displacements associated with a three-phase system.

It is usually convenient to carry out the computations on a single-phase, such as per-phase-Y, line-to-neutral basis, given transformer impedances can then be added

³⁰As there is no neutral connection to carry harmonics of the exciting current, harmonic voltages are produced which significantly distort the transformer voltages.



Figure 2.14

A standard three-phase transformer inside its closure. Observe that while it is three-phase all of its six windings were wound onto the three legs, where the low voltage winding is inside and the high voltage winding is outside.

directly in series with transmission line impedances. The impedances of transmission lines can be referred from one side of the transformer bank to the other by use of the square of the ideal line-to-line voltage ratio of the bank.

In dealing with Y- Δ or Δ -Y banks, all quantities can be referred to the Y-connected side. In dealing with Δ - Δ banks in series with transmission lines, it is convenient to replace the Δ -connected impedances of the transformers by equivalent Y-connected impedances.

A final point to mention is that a balanced Δ -connected circuit of $Z_{\Delta} \Omega \cdot \text{phase}^{-1}$ is **equivalent** to a balanced Y-connected circuit of $Z_Y \Omega \cdot \text{phase}^{-1}$ if

$$Z_Y = \frac{1}{3} Z_{\Delta}$$

A Poly-phase Transformer

2.1 Exercise

Three standard transformers are reconnected Δ - Δ and supplied with power through a 2400-V (line-to-line) three-phase feeder whose reactance is $0.80 \Omega \cdot \text{phase}^{-1}$.

At its sending end, the feeder is connected to the secondary terminals of a three-phase Y- Δ -connected transformer whose rating is:

$$500 \text{ kV} \cdot \text{A}, \quad 24 \text{ kV} : 2,400 \times 10^3 \text{ V} \quad \text{line-to-line}$$

The equivalent series impedance of the sending-end transformer is:

$$0.17 + j0.92 \Omega \cdot \text{phase}^{-1}$$

referred to the 2400-V side. The voltage applied to the primary terminals of the sending-end transformer is 24.0 kV line-to-line.

A three-phase short circuit occurs at the 240-V terminals of the receiving-end transformers.

Using this information, calculate the steady-state short-circuit current in the 2400-V feeder phase wires, in the primary and secondary windings of the receiving-end transformers, and at the 240-V terminals.

The calculations will be made on an equivalent line-to-neutral basis with all quantities referred to the 2400-V feeder. The source voltage then is:

$$\frac{2400}{\sqrt{3}} = 1385 \text{ V} \quad \text{line-to-neutral}$$

From Eq. (2.14), the single-phase-equivalent series impedance of the Δ - Δ transformer seen at its 2400-V side is

$$Z_{eq} = R_{eq} + jX_{eq} = \frac{1.42 + j1.82}{3} = 0.47 + j0.61 \Omega \cdot \text{phase}^{-1}$$

The total series impedance to the short circuit is then the sum of this impedance, that of sending-end transformer and the reactance of the feeder

$$Z_{tot} = (0.47 + j0.61) + (0.17 + j0.92) + j0.80 = 0.64 + j2.33 \Omega \cdot \text{phase}^{-1}$$

which has a magnitude of

$$|Z_{tot}| = 2.42 \Omega \cdot \text{phase}^{-1}$$

Continues on next page →



← Continued from previous page

The magnitude of the phase current in the 2400-V feeder can now simply be calculated as the line-neutral voltage divided by the series impedance

$$\text{Current in 2400-V feeder} = \frac{1385}{2.42} = 572\text{A}$$

and, the winding current in the 2400-V winding of the receiving-end transformer is equal to the phase current divided by $\sqrt{3}$ or

$$\text{Current in 2400-V windings} = \frac{572}{\sqrt{3}} = 330\text{A}$$

while the current in the 240-V windings is 10 times this value

$$\text{Current in 240-V windings} = 10 \times 330 = 3300\text{A}$$

Finally, the phase current at the 240-V terminals into the short circuit is given by

$$\text{Current at the 240-V terminals} = 3300\sqrt{3} = 5720\text{A}$$

Note of course that this same result could have been computed simply by recognizing that the turns ratio of the Δ - Δ transformer bank is equal to 10:1 and hence, under balanced-three-phase conditions, the phase current on the low voltage side will be 10 times that on the high-voltage side. ■

2.4.3 Voltage and Current Transformers

Transformers are often used in instrumentation applications to **match the magnitude of a voltage** or **current** to the range of a meter or other instrumentation. As an example, most 60 Hz power-systems' instrumentation is based upon voltages in the range of 0 V to 10 V RMS and currents in the range of 0 A to 5 A RMS. Since power system voltages range up to 765 kV line-to-line and currents can be 10's of kA, some method of supplying an accurate, low-level representation of these signals to the instrumentation is needed.

A common technique is through the use of specialized transformers known as **potential transformers** (PT)s and **current transformers** (CT)s. If constructed with a turns ratio of $N_1 : N_2$, an ideal potential transformer would have a secondary voltage equal in magnitude to N_2/N_1 times that of the primary and identical in phase.

Similarly, an **ideal current transformer** would have a secondary output current equal to N_1/N_2 times the current input to the primary, again identical in phase.

Potential and current transformers³¹ are designed to be ideal transformers as closely as possible.

³¹also referred to as instrumentation transformers.

The equivalent circuit of **Fig. 2.15** shows a transformer loaded with an impedance $Z_b = R_b + jX_b$ at its secondary. For the sake of this discussion, the core-loss resistance R_c has been neglected.³² Following conventional terminology, the load on an instrumentation transformer is frequently referred to as the **burden** on that transformer, hence the subscript b . To simplify our discussion, we have chosen to refer all the secondary quantities to the primary side of the ideal transformer.

³²However, if there is a need, the analysis presented here can be easily expanded to include its effect.

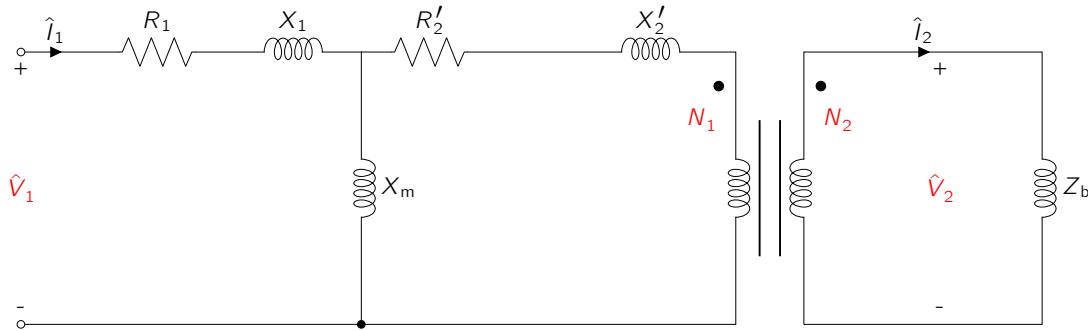


Figure 2.15

Equivalent circuit for an instrumentation transformer.

Voltage Transformer

Ideally it should accurately measure voltage while appearing as an open circuit to the system under measurement, meaning it should be drawing negligible current and power. Therefore, its load impedance should be **large**.

To start, let us assume the transformer secondary is open-circuited, i.e., $|Z_b| = \infty$. In this case:

$$\frac{\hat{V}_2}{\hat{V}_1} = \left(\frac{N_2}{N_1} \right) \frac{jX_m}{R_1 + j(X_1 + X_m)} \quad (2.14)$$

As we see, a potential transformer with an open-circuited secondary has an inherent error, in both magnitude and phase, due to the voltage drop of the **magnetizing current** through the primary resistance and leakage reactance.

The difference of magnitude is high enough to make primary resistance and leakage reactance small compared to the magnetizing reactance, this inherent error can be made quite small. The situation is worsened by the presence of a finite burden. Including the effect of the burden impedance, Eq. (2.14) becomes:³³

$$\frac{\hat{V}_2}{\hat{V}_1} = \left(\frac{N_2}{N_1} \right) \frac{Z_{eq} Z'_b}{(R_1 + jX_1)(Z_{eq} + Z'_b + R'_2 + jX'_2)} \quad \text{where} \quad Z_{eq} = \frac{jX_m(R_1 + jX_1)}{R_1 + j(X_m + X_1)}$$

$$Z'_b = \left(\frac{N_1}{N_2} \right)^2 Z_b$$

is the burden impedance referred to the transformer primary.

From these equations, it can be seen that the characteristics of an accurate potential transformer include a large magnetizing reactance³⁴ and relatively small winding resistances and leakage reactances. Finally the burden impedance must be kept above a minimum value to avoid introducing excessive errors in the magnitude and phase angle of the measured voltage.

Current Transformer



Figure 2.16

A 120:120 instrument isolation transformer showing two polarity marking conventions.

³³Please observe the burden impedance has been referred to the primary.

³⁴more accurately, a large exciting impedance since the effects of core loss, although neglected in the analysis presented here, must also be minimised.

An ideal current transformer would accurately measure voltage while appearing as a short circuit to the system under measurement.³⁵ Therefore, its load impedance should be “small” in a sense we will now quantify.

Let's start with the assumption that the transformer secondary is short-circuited, meaning $|Z_b| = x0$. In this case we can write that:

$$\frac{\hat{I}_2}{\hat{I}_1} = \left(\frac{N_1}{N_2} \right) \frac{jX_m}{R'_2 + j(X'_2 + X_m)} \quad (2.15)$$

In a fashion quite analogous to that of a potential transformer, Eq. (2.15) shows that a current transformer with a shorted secondary has an inherent error in both magnitude and phase, due to the fact that some of the primary current is shunted through the magnetizing reactance and does not reach the secondary.

To the extent that the magnetizing reactance can be made large in comparison to the secondary resistance and leakage reactance, this error can be made quite small.

A finite burden appears in series with the secondary impedance and increases the error. Including the effect of the burden impedance which rewrites as Eq. (2.15):

$$\frac{\hat{I}_2}{\hat{I}_1} = \left(\frac{N_1}{N_2} \right) \frac{jX_m}{Z'_b + R'_2 + j(X'_2 + X_m)}$$

From these equations, it can be seen that an accurate current transformer has a large magnetizing impedance and relatively small winding resistances and leakage reactances. In addition, the burden impedance on a current transformer must be kept below a maximum value to avoid introducing excessive additional magnitude and phase errors in the measured current.

³⁵i.e., developing negligible voltage drop and drawing negligible power.



Figure 2.17

A CT for operation on a 110 kV grid.

References



Part II

Rotating Machines

Part Contents

Invention is the most important product of man's creative brain. The ultimate purpose is the complete mastery of mind over the material world, the harnessing of human nature to human needs.

(Nikola Tesla, My Inventions)



Part III

Machine as Drives

Part Contents

We are becoming the servants in thought, as in action, of the machine we have created to serve us.

(John Kenneth Galbraith (2007), The new industrial state.)



Part IV

Appendix

Part Contents

Acronyms	55
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The end of a melody is not its goal: but nonetheless, had the melody not reached its end it would not have reached its goal either.

A parable.

(Friedrich Nietzsche)

Acronyms

		A
AC	An electric current that periodically reverses direction and changes its magnitude continuously with time, in contrast to direct current (DC), which flows only in one direction.	
		D
DC	One-directional flow of electric charge. Direct current may flow through a conductor such as a wire, but can also flow through semiconductors, insulators, or even through a vacuum as in electron or ion beams. The electric current flows in a constant direction, distinguishing it from alternating current (AC). A term formerly used for this type of current was galvanic current.	
		E
EMF	An energy transfer to an electric circuit per unit of electric charge, measured in volts. Devices called electrical transducers provide an EMF by converting other forms of energy into electrical energy. Other types of electrical equipment also produce an emf, such as batteries, which convert chemical energy, and generators, which convert mechanical energy. This energy conversion is achieved by physical forces applying physical work on electric charges. However, electromotive force itself is not a physical force	
		L
LHS	Left Hand Side	
		M
MMF	The magnetic pressure which drives magnetic flux in a magnetic circuit, and is the work done to move a unit magnetic pole.	
		P
PM	An object made from a material that is magnetized and creates its own persistent magnetic field	
		R
RHS	Right Hand Side	



RMS Equivalent direct current (DC) voltage of an alternating current (AC) source.



The document is designed with no intention of publication and has only been designed for education purposes.

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